

Modelling Phenological Shifts in Similipal National Park Using MODIS Satellite Data (2000–2024)

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📍 SIMILIPAL NATIONAL PARK, ODISHA, INDIA

📅 2000–2024



Why Study Phenological Shifts?

What is Phenology?

It is the timing of seasonal vegetation events — like when leaves appear or when they fall.

Climate Change Impact

Climate change is altering these timings globally.

Key Questions We Are Trying to Answer

- 1 Is vegetation starting to grow **earlier** over the years?
- 2 Is the growing season becoming **longer**?
- 3 Are these shifts linked to **monsoon or climate patterns**?



Understanding these shifts is critical for forest management, carbon cycle modelling, and biodiversity conservation.

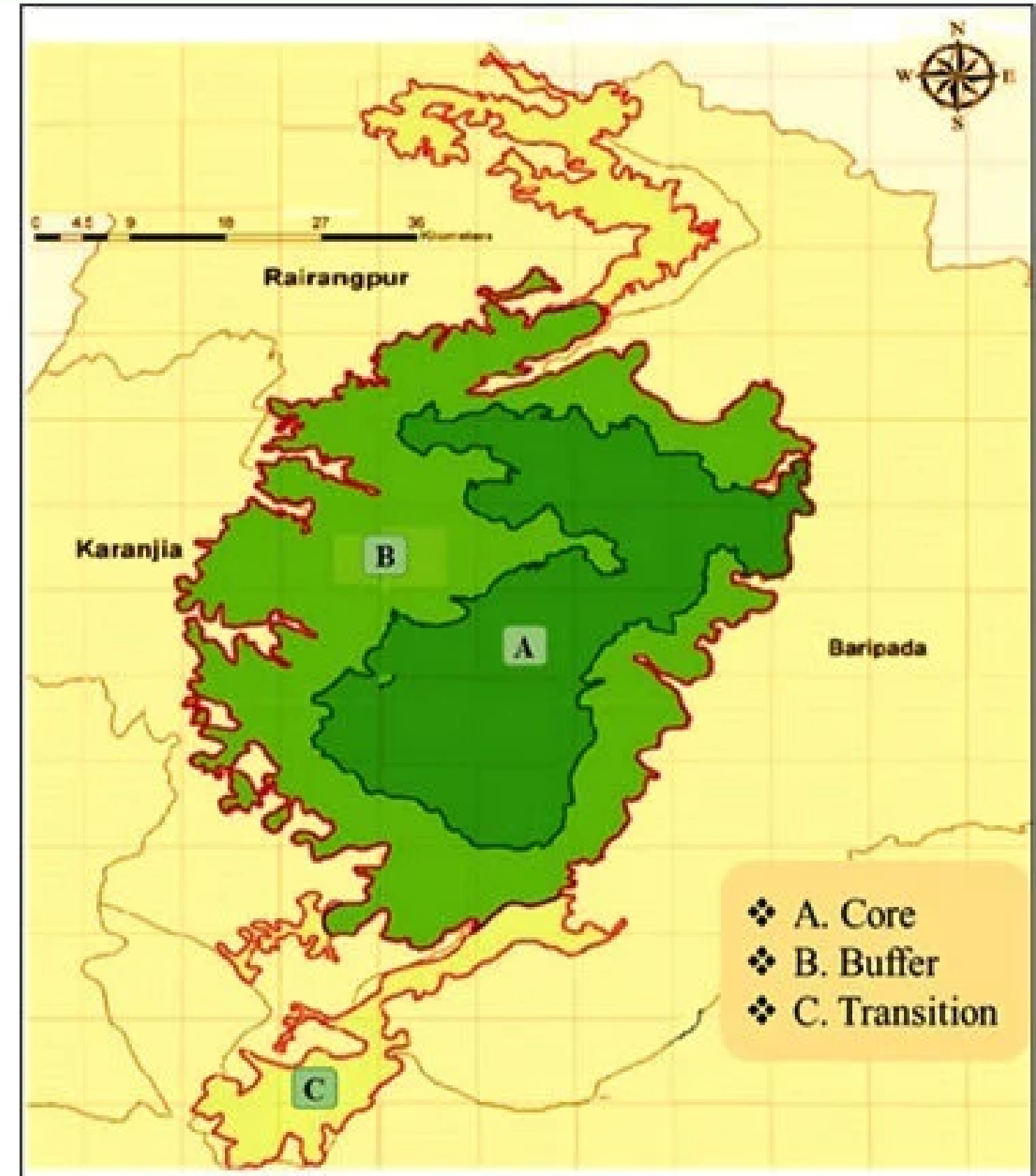
Study Area — Similipal National Park, Odisha

Location & Characteristics

- **Location:** Mayurbhanj District, Odisha, Eastern India
- **Area:** ~850 km² — one of India's largest national parks
- **Forest type:** Tropical Moist Deciduous Forest

Why We Chose This Area

- Large continuous forest (suitable for MODIS 250m/500m data)
- Strong monsoon-driven seasonal vegetation
- Minimal urban disturbance
- 25-year consistent satellite data availability



Literature Review — Yu et al. (2017)

"MONITORING THE LONG-TERM VEGETATION PHENOLOGY CHANGE DYNAMICS IN NORTHEAST CHINA" — SCIENTIFIC REPORTS

What We Learned

- Used the **GIMMS NDVI3g** dataset derived from **AVHRR** satellite sensors (1982–2015) to detect long-term phenological shifts in vegetation.
- Applied **Savitzky-Golay smoothing** to remove noise from satellite data
- Used the **amplitude threshold method** to extract Start, End, and Length of Season (SOS, EOS, LOS)
- Found shifts are linked to temperature and precipitation

How We Adapted It

- Same smoothing + threshold approach
- Modified for **tropical deciduous forest** (vs temperate regions)
- Used a **March-to-March phenological year** instead of a calendar year

Why a "March-to-March" Phenological Year?

The Challenge with a Normal Calendar (Jan–Dec)

In tropical monsoon deciduous forests like Similipal, the leaf-fall and leaf-flush (growing season) bridge two calendar years. Breaking the year at December 31st cuts the continuous ecological season exactly in half, splitting a single biological cycle.

The Solution (March to March)

Pre-Monsoon Start

Trees in this region often shed leaves and begin preparing for new leaf flush around March/April, induced by pre-monsoon heat.

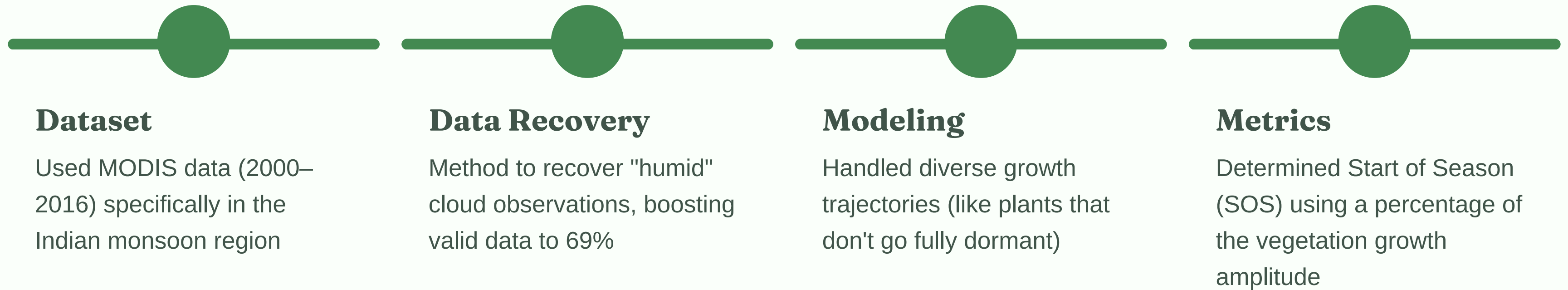
Full Cycle Integrity

A March-to-March timeline perfectly captures the entire biological cycle — leaf out, peak greenness (monsoon), and senescence (winter) — inside one continuous sequence.

Better Detection

This allows mathematical algorithms to smoothly find the "Start" and "End" of the season without artificial breaks.

Literature Review — Shang et al. (2018)



How We Adapted It

- Threshold Standardization for our regions
- Learned how Indian forests vary seasonally
- Improved our Trajectory Modeling

Data Extraction: High-Resolution Pixel-Wise Approach

Objective

Combine multiple satellite sources into a single, comprehensive dataset. Capture conditions down to the **pixel level (500m resolution)**. No regional averaging — we track *every single pixel* from 2001 to 2024.

MODIS Versions - Metrics Extracted

MOD09A1 (Version 6.1): NDVI, EVI, NDMI, NDWI

MOD11A2 Version 6 and MOD11A2 Version 6.1 - LST_Day_C, LST_Night_C, LST_DTR

MOD16A2GF Version 6.1: ET_mm8day, LE_MJm2day, PET_mm8day, ET_stress

MOD17A2H Version 6 and MOD17A2H Version 6.1: GPP_gCm2_8day, PsnNet_gCm2_8day

CHIRPS Daily Precipitation: rainfall_mm_8day

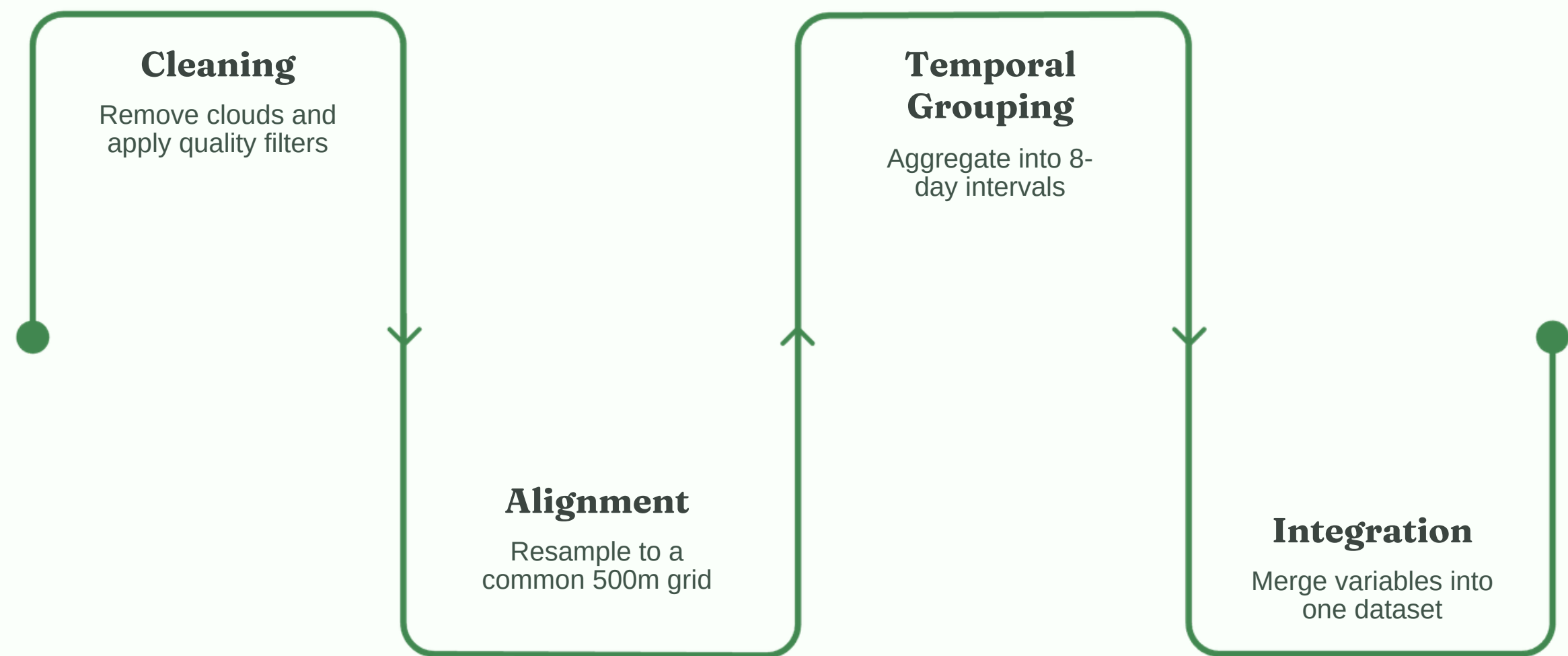
Data Sources Collected per Pixel

Vegetation NDVI & EVI (Health), NDWI & NDMI (Moisture)	Temperature LST Day, Night, and Range
Water Stress Evapotranspiration (ET) & Drought stress	Rainfall 8-day CHIRPS totals
Productivity Photosynthesis (GPP)	

 Combine multiple satellite sources into a single, comprehensive dataset. Capture conditions down to the **pixel level (500m resolution)**. No regional averaging — we track *every single pixel* from 2001 to 2024.

Data Integration and Final Structure

Methodology



Each step ensures data quality, spatial consistency, temporal alignment, and a unified structure ready for phenological analysis.

Final Output Structure

Format One CSV file per year (to handle massive data size)	Rows Each row is one pixel at one time step (~250,000 rows/year)	Result A pixel-perfect timeline ready for phenology and climate analysis
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Classifying the Forest: Wet vs. Dry Zones



Our Adaptation: Instead of K-means, we built a deterministic index tailored for forest moisture tracking.

Scientific Inspiration

We referred to a clustering approach (*Moussaid et al., 2024*) that used unsupervised K-means clustering on satellite water indices (NDVI, MNDWI) for mapping agricultural land.

Methodology Details

Per-Pixel Long-term Averages

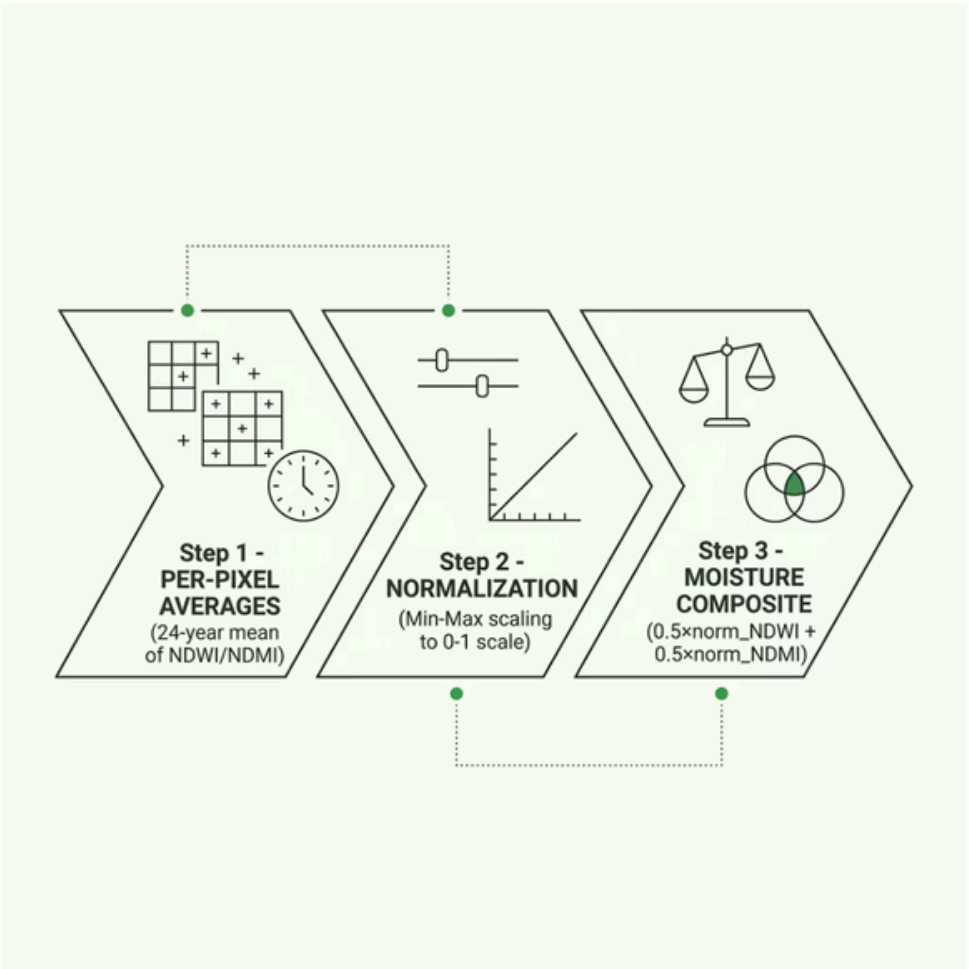
Calculated the 24-year mean for two moisture indices per pixel: **NDWI** (Normalized Difference Water Index) — measures liquid water in vegetation; and **NDMI** (Normalized Difference Moisture Index) — captures overall canopy & soil moisture stress.

Normalization

Both indices were Min-Max normalized to a 0–1 scale so they are perfectly comparable.

Moisture Composite

Combined with equal weight: **Moisture Composite** = $(0.5 \times \text{norm_NDWI}) + (0.5 \times \text{norm_NDMI})$



Assigning the Wet and Dry Zones

Thresholding the Moisture Composite: Based on the Moisture Composite scores of all 10,000+ pixels, we split the forest statistically by calculating the exact 50th percentile (the median score).

● **Wet Zone (~49.5%)**

Pixels **above** the median threshold. These areas retain higher moisture year-round (e.g., valleys, deeper soils).

● **Dry Zone (~49.5%)**

Pixels **below** the median threshold. These areas dry out faster (e.g., ridges, rocky slopes, degraded edges).

● **Buffer Zone (1%)**

A narrow 1% margin exactly on the median was labeled "**Others**" to cleanly separate the core Wet and Dry pixels from border ambiguity.

□ This allows us to independently track if the Dry areas are shifting differently than the Wet areas under climate stress.

Extracting Phenological Metrics

Once the NDVI curves were smoothed using Savitzky-Golay filters, we extracted these core metrics per pixel, per year:

SOS — Start of Season

The day of the year when rapid leaf growth begins.

EOS — End of Season

The day when leaves significantly fall or brown.

LOS — Length of Season

Total days the forest was "active" (EOS – SOS).

Peak NDVI

The maximum greenness achieved during the monsoon.

Integrated NDVI

The total area under the greenness curve, used as a proxy for total annual productivity.

Setting the Phenology Amplitude Thresholds

How is SOS/EOS mathematically defined? We used the **Amplitude-Fraction Method**. We find the lowest "baseline" greenness and the "peak" greenness. The difference is the **Seasonal Amplitude**. SOS is the day when NDVI crosses a specific percentage of this amplitude on the way up. EOS is when it crosses on the way down.



Our Custom Thresholds

Dry Zone: 20% (0.20)

Wet Zone: 35% (0.35)

Why Different Thresholds for Wet and Dry Zones?

One universal threshold fails when dealing with heterogeneous forests. Based on literature reviews of standardised phenology methods, thresholds must match the ecosystem's baseline density.

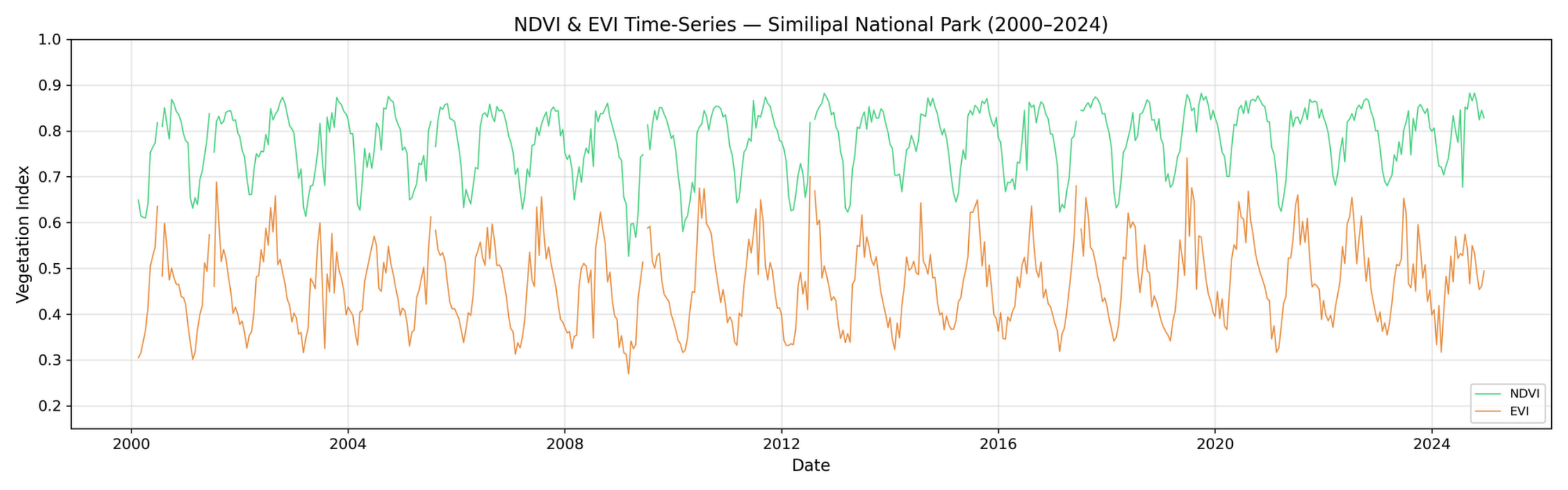
Dry Zones — 20%

These areas have sparser canopies and lower background greenness. Even a small 20% rise above baseline reliably indicates that new seasonal growth (flush) has truly started.

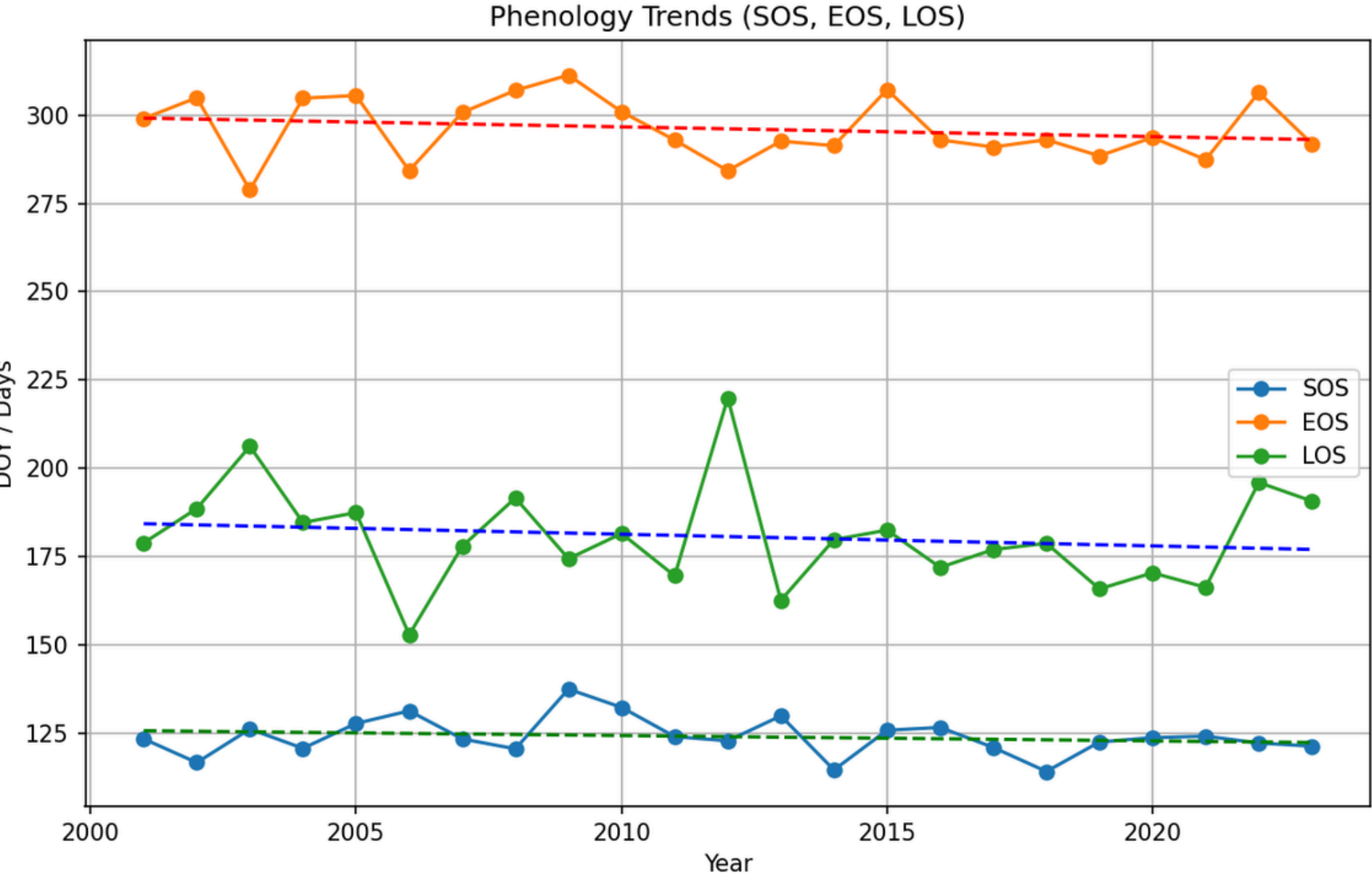
Wet Zones — 35%

These areas hold dense, semi-evergreen vegetation. The baseline is already high and noisy. We require a stricter 35% jump to distinguish genuine new seasonal growth from existing background foliage fluctuations.

NDVI & EVI Time-Series



SOS, EOS, and LOS Trends (2001–2024)



Key Observations

The Central Result

This directly answers our main research question.

Start of Season (SOS)

Shows a clear upward trend line (delaying) in certain zones over time.

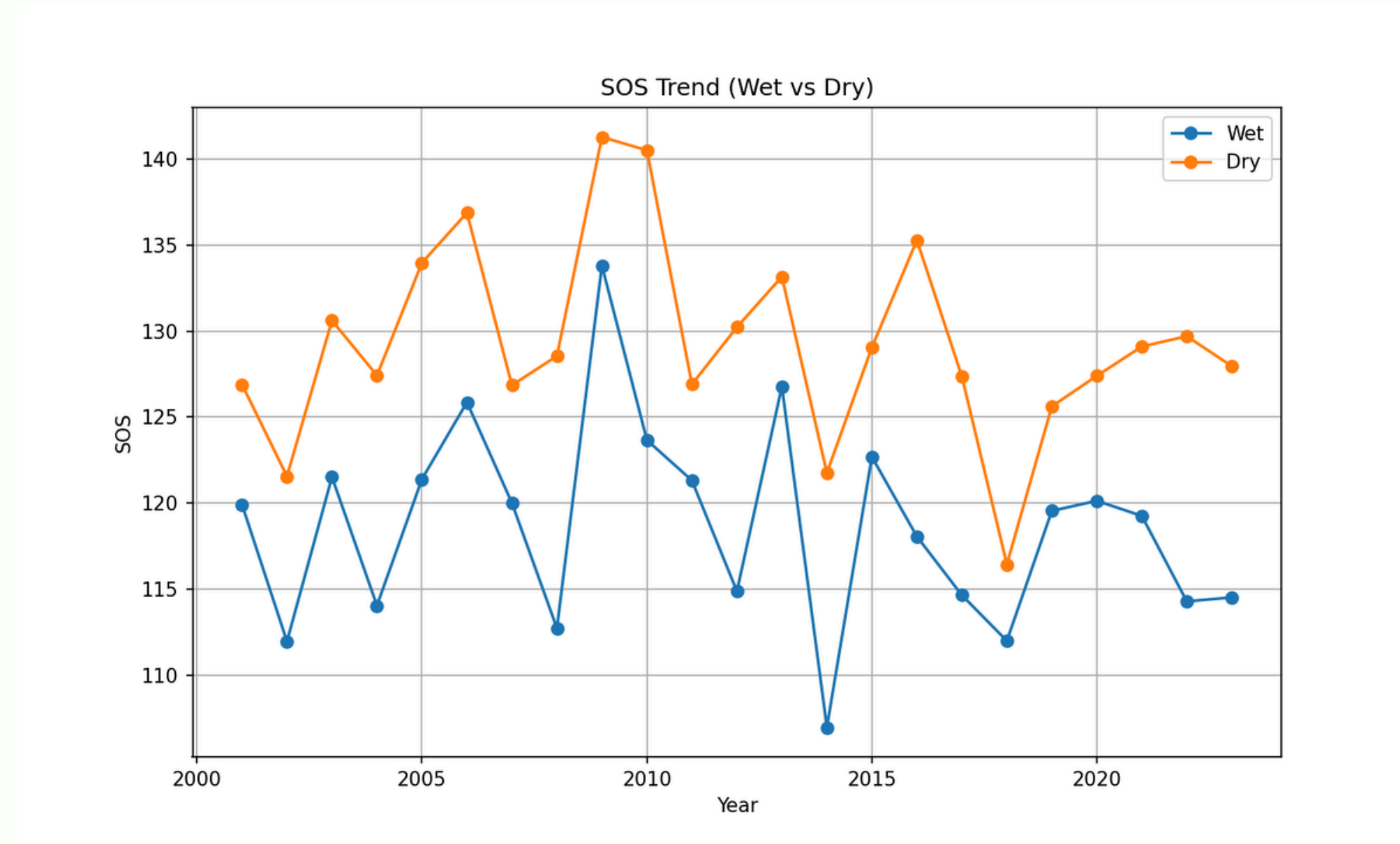
Length of Season (LOS)

Because SOS is shifting, the total active growing days (LOS) is changing.

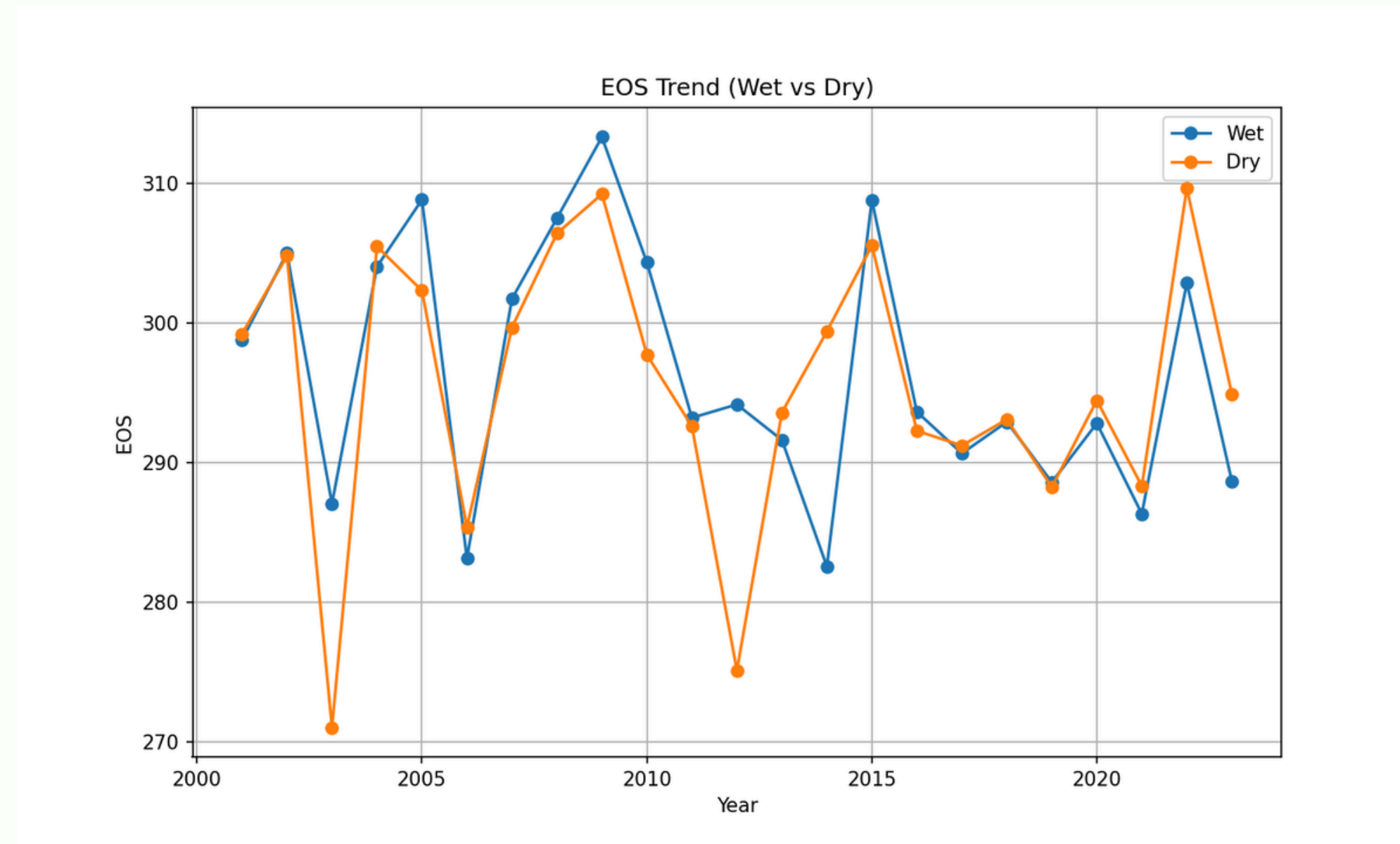
Conclusion

Phenological timing is not stable; there is a distinct, measurable multi-decadal shift occurring in Similipal.

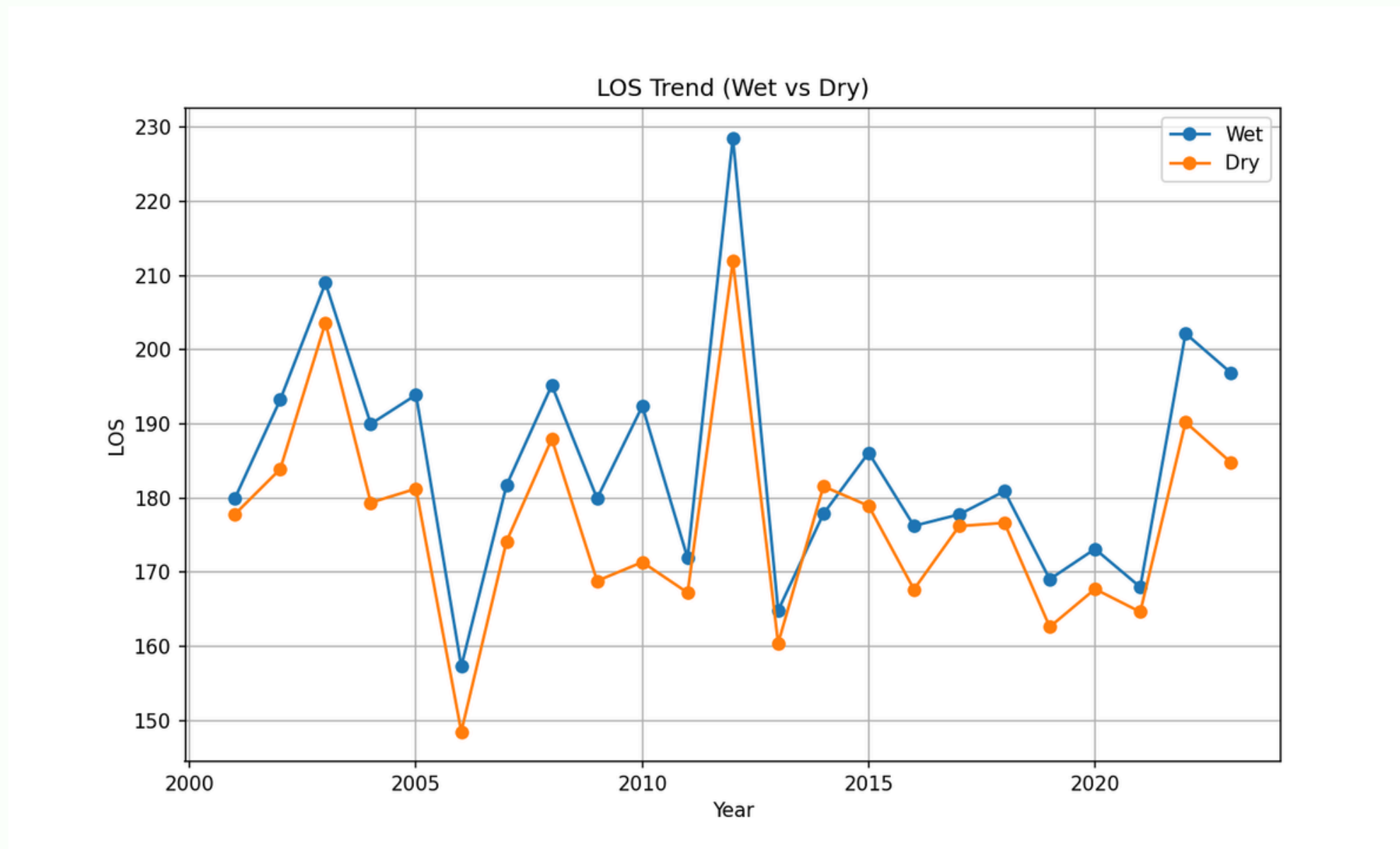
SOS Time-Series



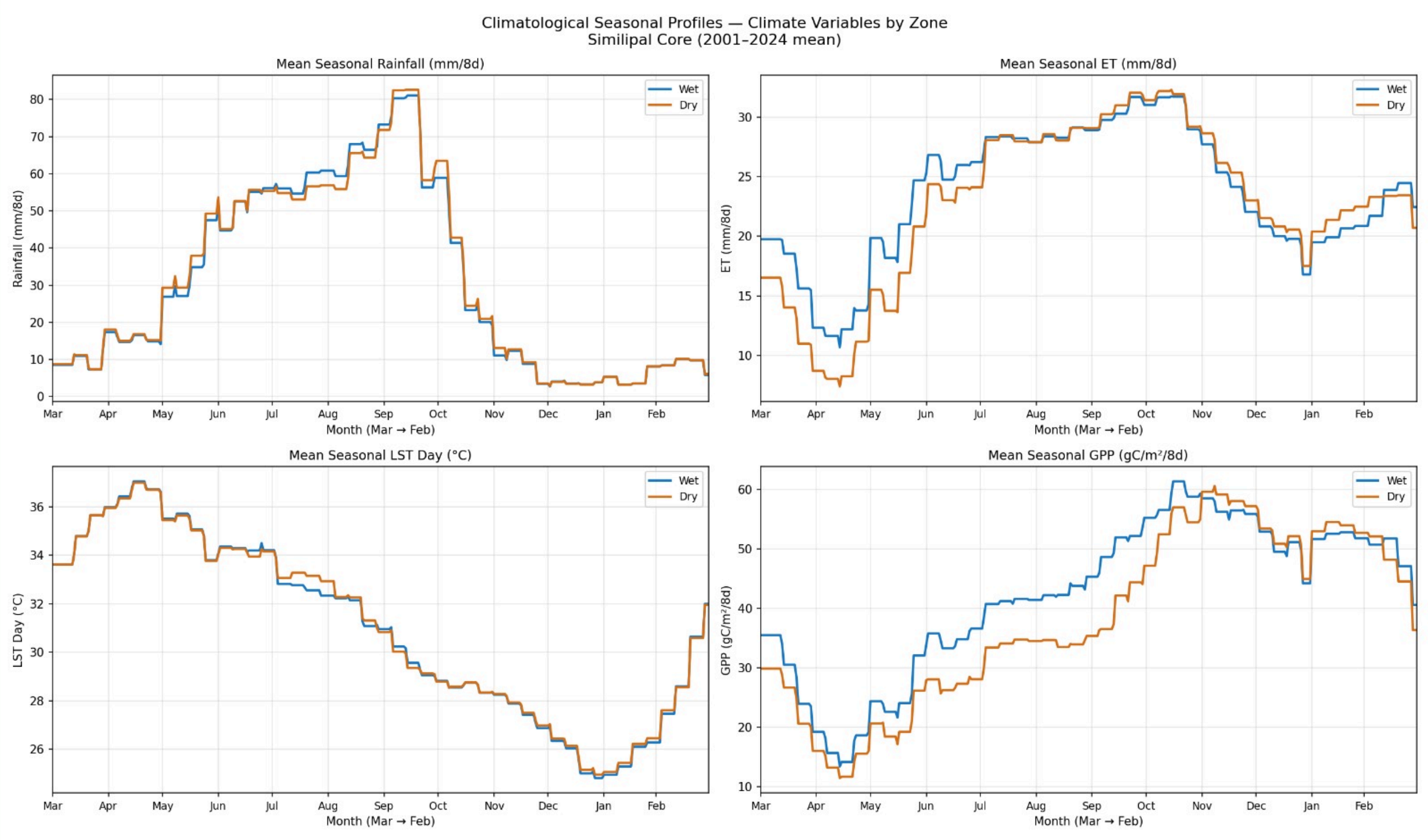
EOS Time Series



LOS Time-Series

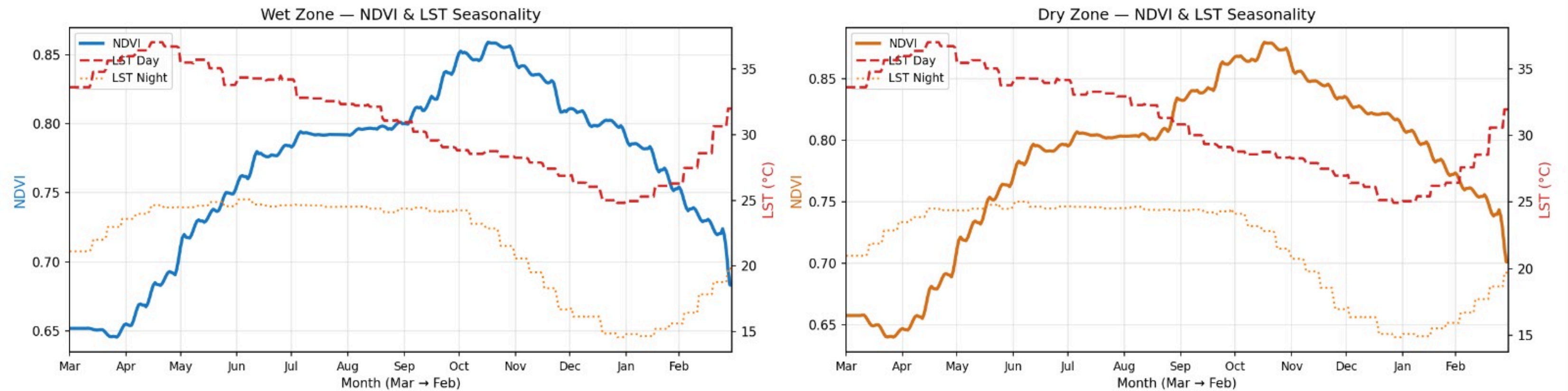


Climatological Seasonal Profiles



NDVI Vs LST

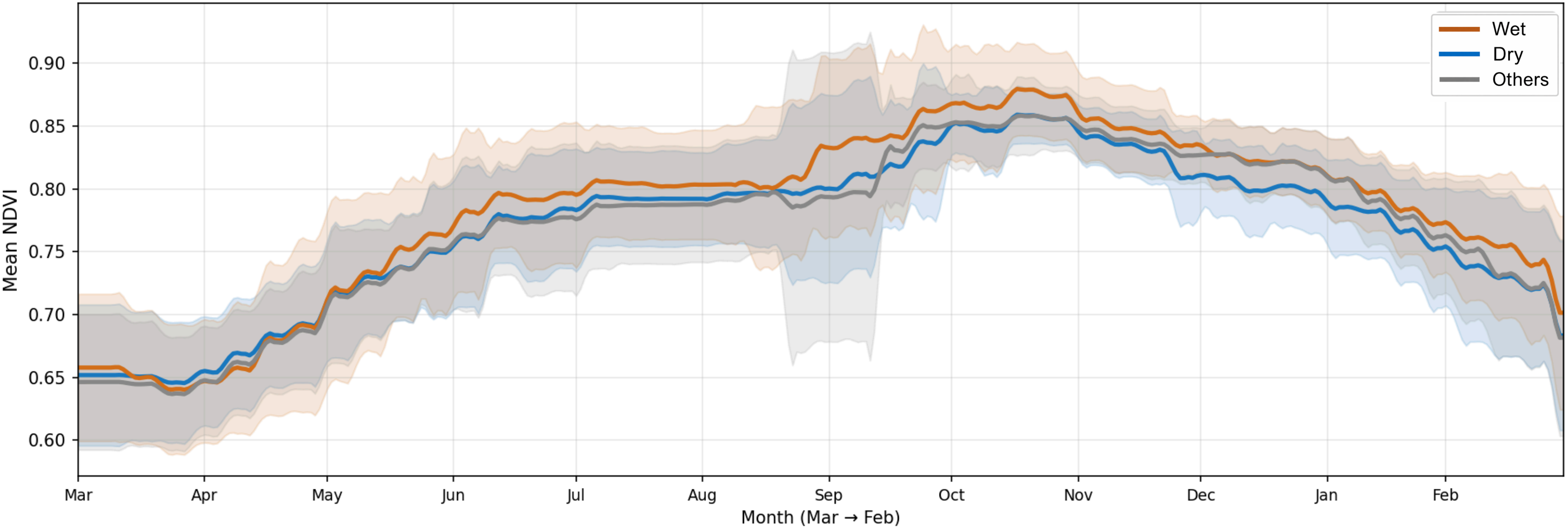
NDVI vs Land Surface Temperature Seasonal Profile
Similipal Core (2001-2024 mean)



The Baseline: 24-Year Mean NDVI (Wet vs. Dry)

— Dry Zone NDVI — Wet Zone NDVI

Long-term Mean Seasonal NDVI Profile — Similipal Core (2001-2024)
Shaded region = ± 1 SD of inter-annual variability



The Baseline: 24-Year Mean NDVI (Wet vs. Dry)

Key Observations

Ecological Context

This curve represents the average growing season over 24 years.

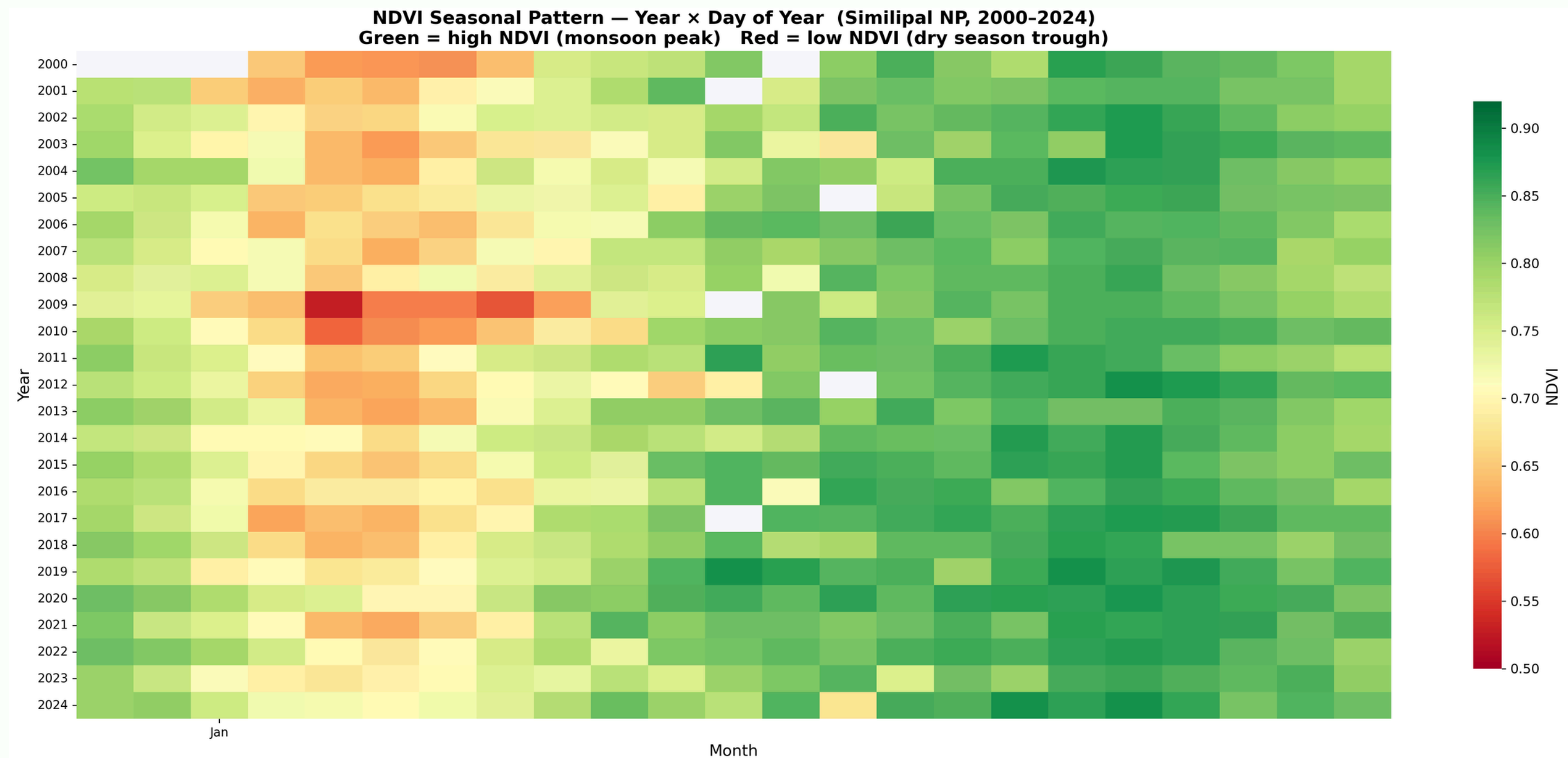
Zone Differences

Wet zones maintain a noticeably higher greenness (NDVI) year-round compared to Dry zones.

Variability

The shaded regions (± 1 Standard Deviation) show the natural inter-annual variation—notice how the start of the season (April–May) has the highest uncertainty, meaning its timing is highly sensitive to annual weather.

Visualising 24 Years of Greenness



Visualising 24 Years of Greenness

Emergent Patterns

A powerful "phenolandscape" showing Year (Y-axis) versus Month (X-axis).

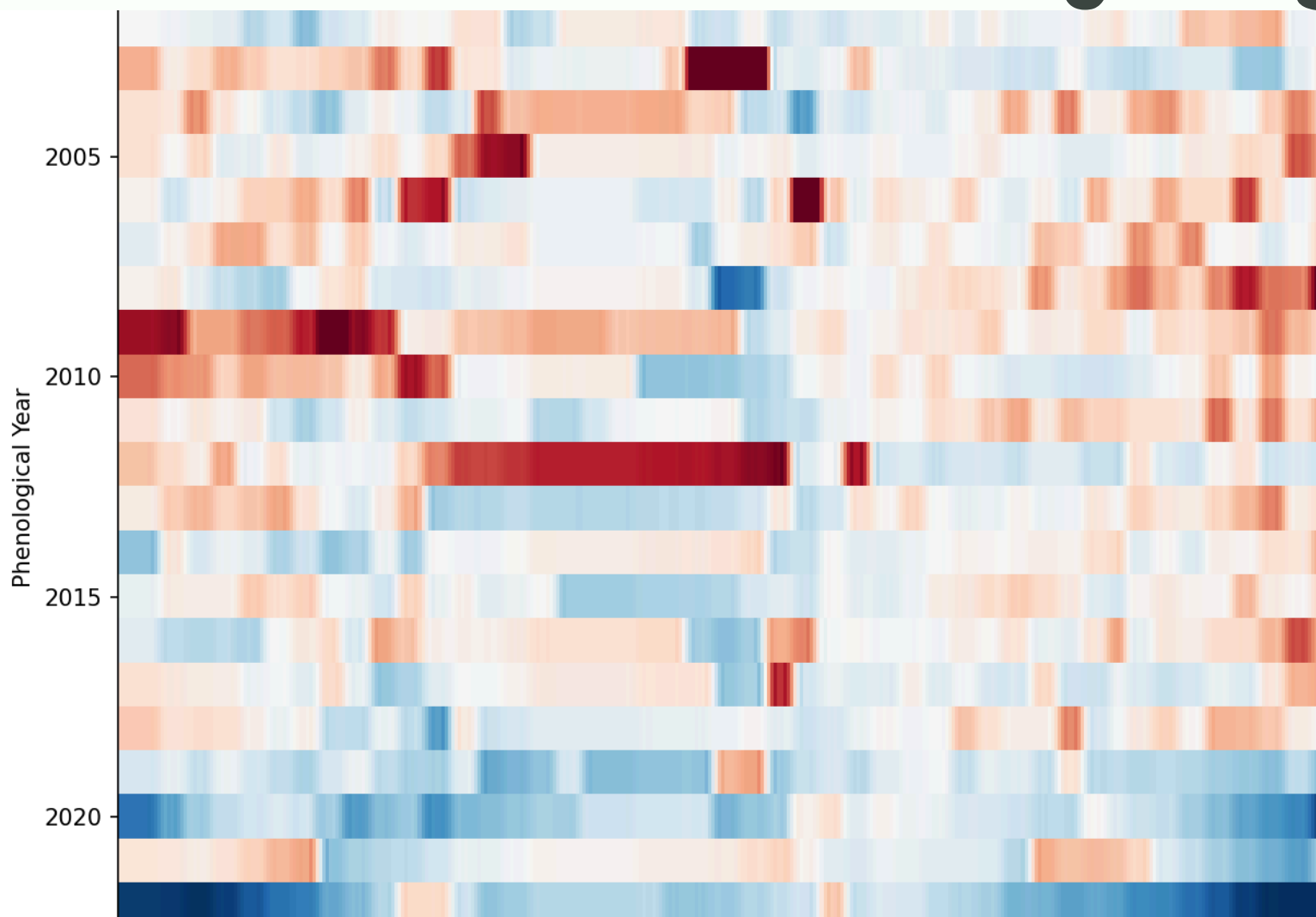
Greenness Shifts

Notice visually how the intense green band (monsoon peak) shifts slightly in width and intensity across different years.

Consistency

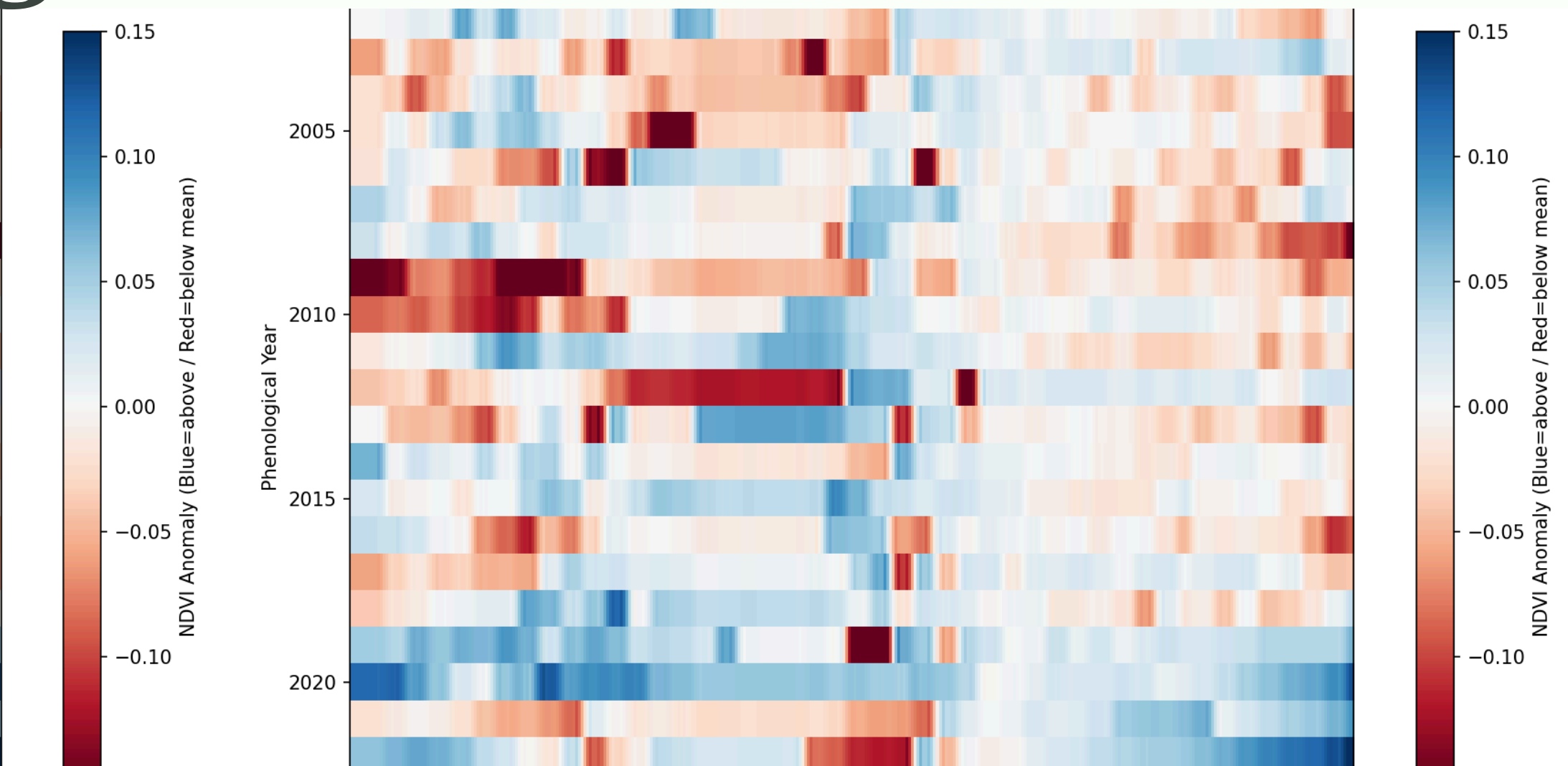
Allows us to instantly spot years with delayed green-up or prolonged dry seasons across the two decades, proving that the forest canopy is highly dynamic, not static.

Identifying Anomalous Years



Tracking Extremes

Blue indicates greener-than-average, and Red indicates browner/drier-than-average periods.



Climate Linkage

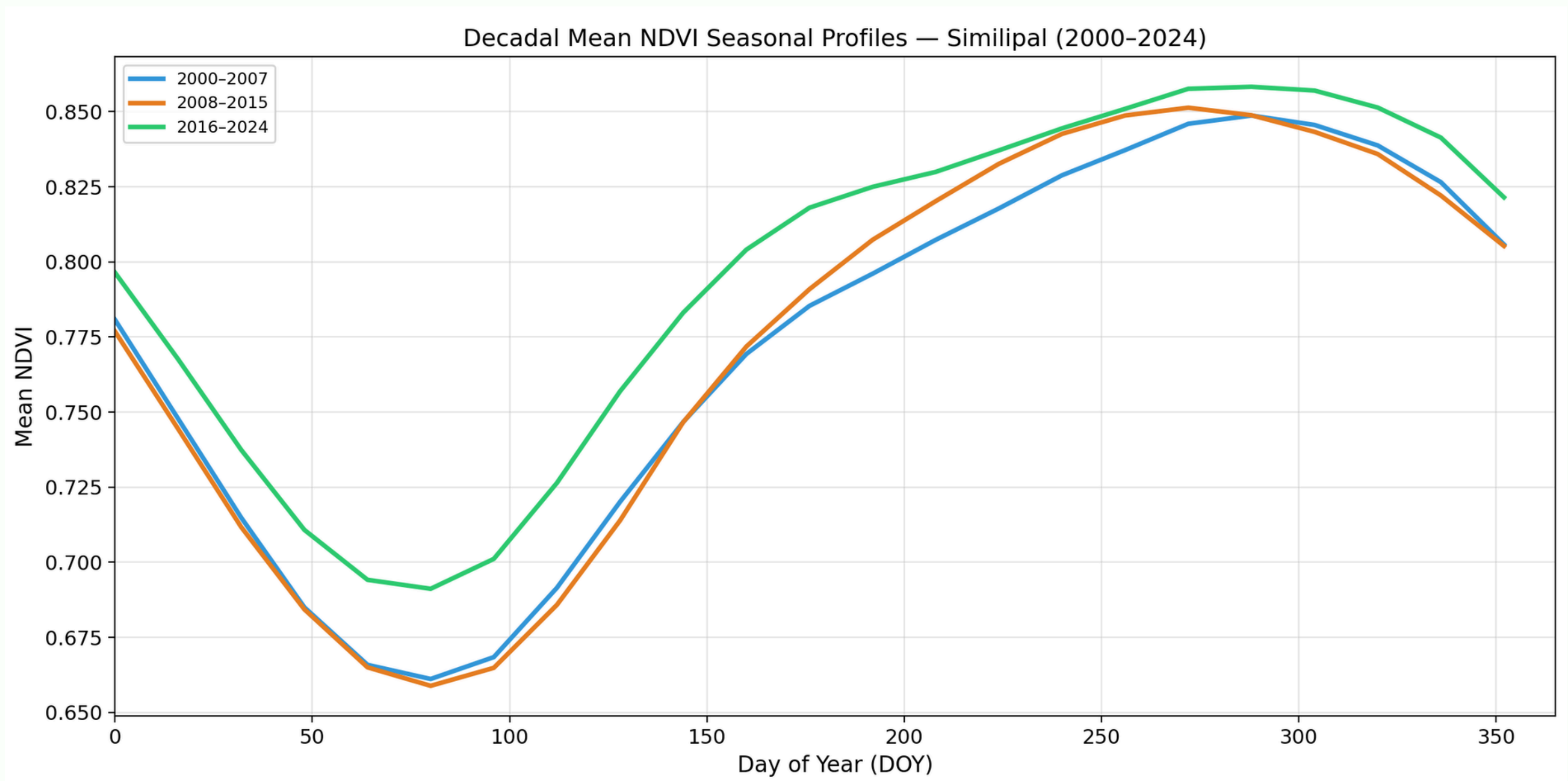
We can directly spot massive stress years (heavy red banding) which align with known historic droughts or delayed monsoons in Odisha.

Zone Resilience

Wet zones often show different anomaly intensities compared to Dry zones, proving they respond differently to the same regional climate shocks.

Decadal Shift Comparison

SOS Shift (Days Delayed) LOS Change (Days)



SOS Shift (Days Delayed)

Key Observations

Time Blocks

Breaking the 24 years into three distinct blocks (2001–07, 2008–15, 2016–24).

Pacing of Shifts

Clearly shows that the shift isn't just a slow straight line. We can observe if recent years (2021–2024) have accelerated phenological delays or extensions compared to the 2000s.

Narrative Impact

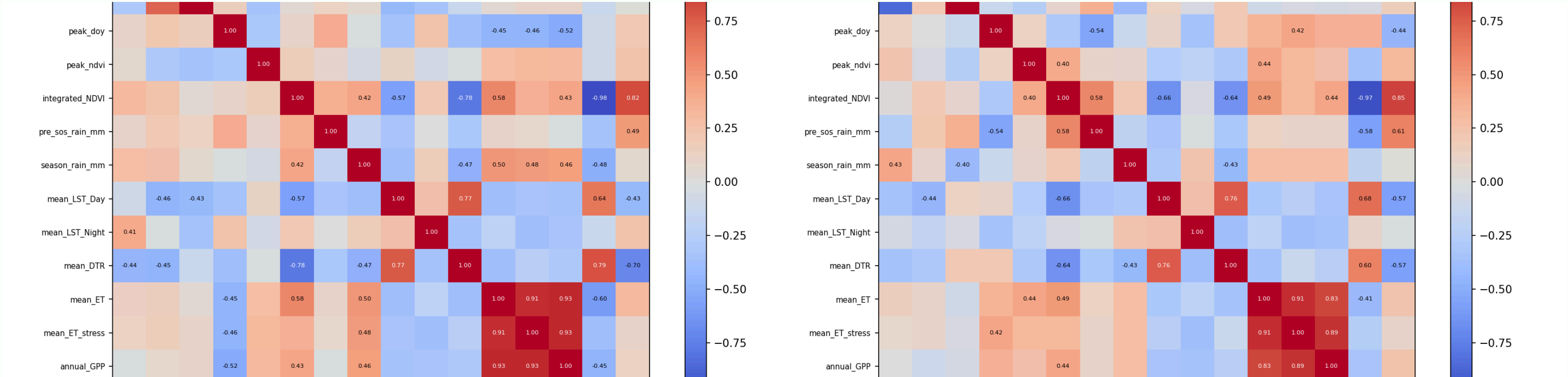
This is critical for the climate-change narrative, showing how recent warming extremes have intensified the biological response.

Decadal
Shift
Comparison

LOS Change (Days)

Correlation Between Phenology and Climate

Key Observations



System Understanding

We tested phenological dates against LST (Temperature), Rainfall, and ET (Evapotranspiration).

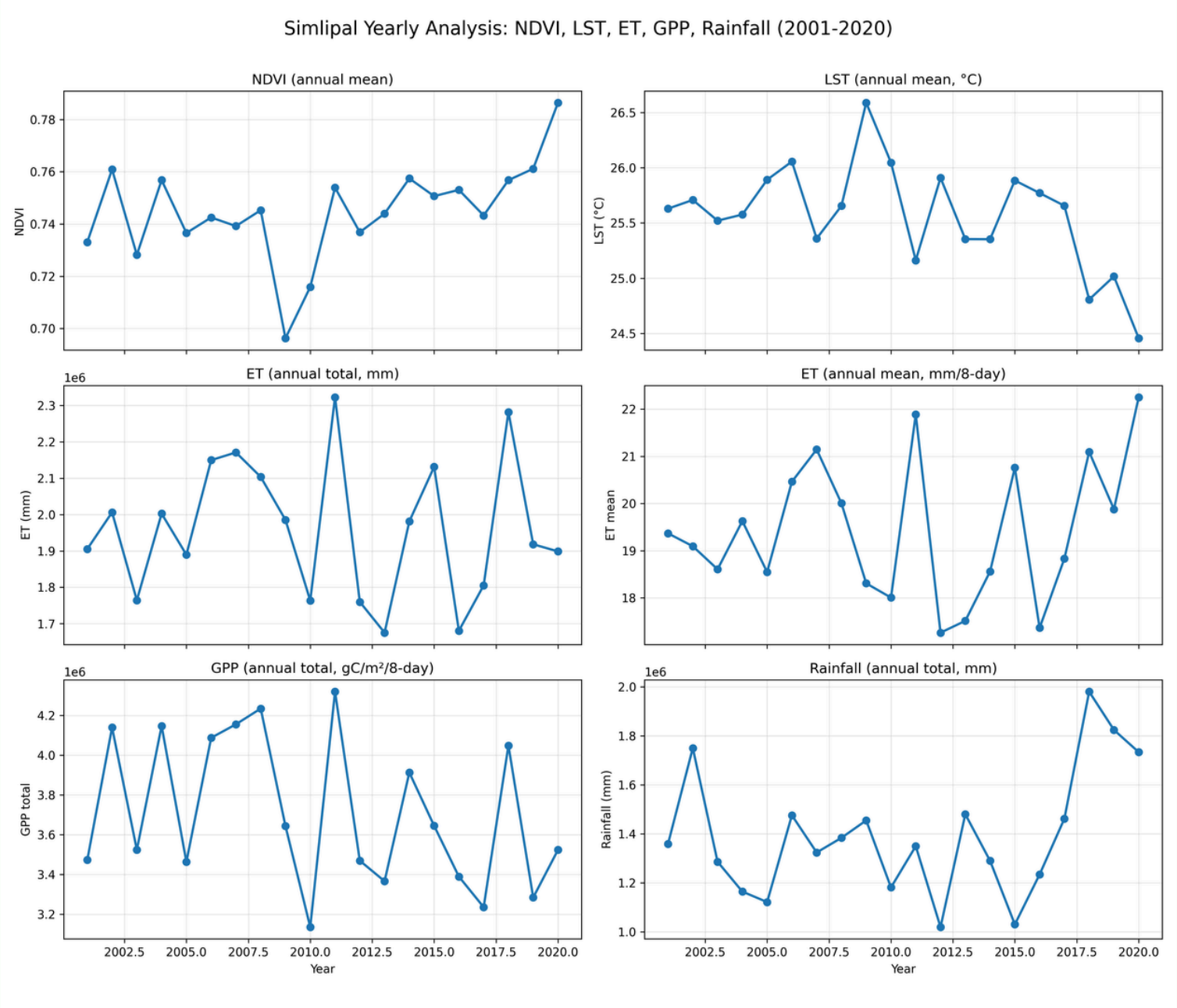
Strong Linkages

Deep red and deep blue squares indicate strong mathematical correlations.

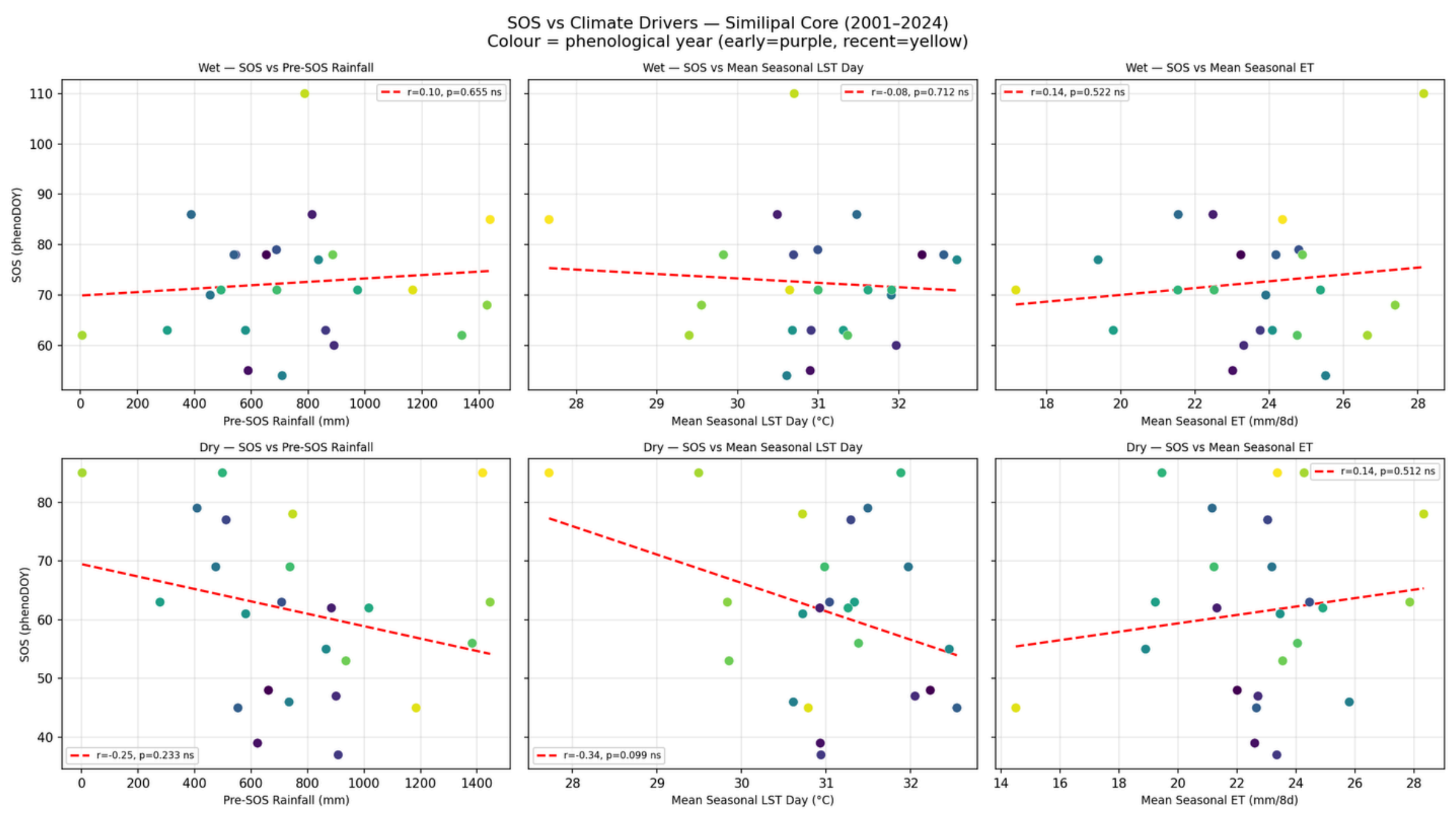
Insight

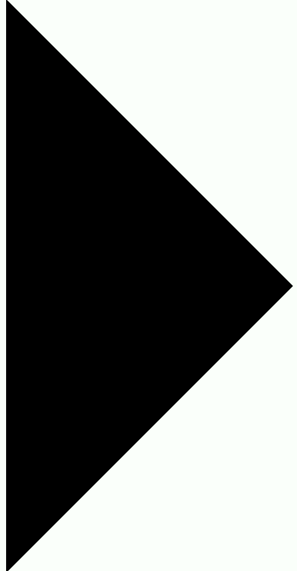
SOS and EOS do not shift randomly; they are intimately tied to pre-monsoon heat and monsoon rainfall totals, proving that climate is the primary driver.

NDVI, LST, ET, Rainfall, GPP Analysis



Attributing Shifts to Climate Forcing



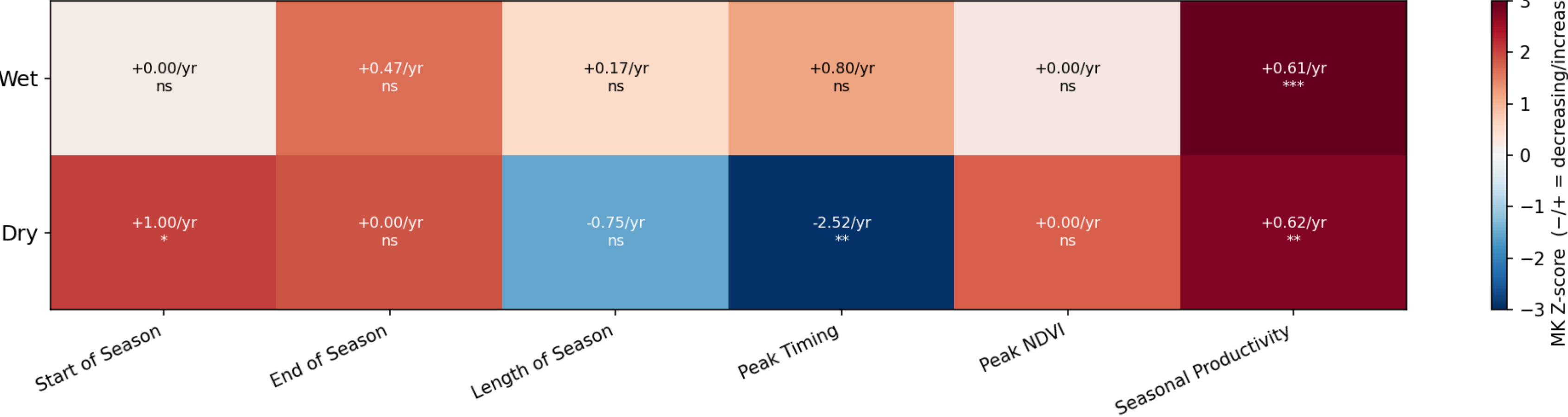


Attributing Shifts to Climate Forcing

Scatter plots plotting Start of Season (SOS) directly against pre-season rainfall and temperatures reveal direct cause-and-effect relationships. Regression lines show exactly how sensitive the forest is: e.g., "For every 1°C increase in pre-monsoon LST, the start of the season shifts by X days." This solidly attributes the phenological shifts to actual climate-forcing variables rather than internal forest dynamics alone.

Mann-Kendall Trend Summary

Mann-Kendall Trend Summary — Similipal Core Area (2001-2024)
Cell text: Sen's slope / significance stars (* p<0.05, ** p<0.01, *** p<0.001)



Mann-Kendall Trend Summary

The Absolute Proof

Summarises the Mann-Kendall Z-scores and Sen's slope for every variable across all zones. Asterisks indicate statistical significance ($p < 0.05$, 0.01, 0.001).

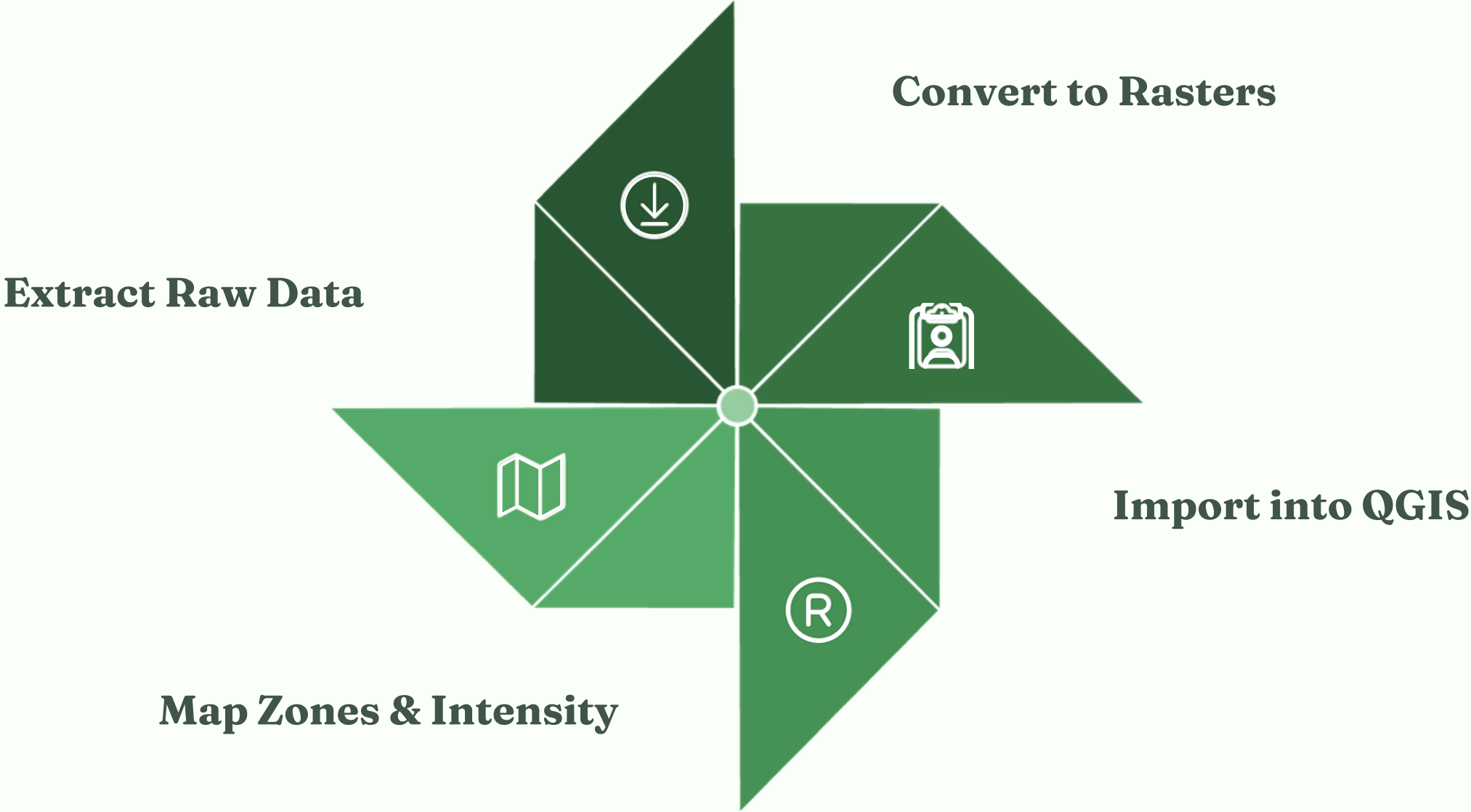
Final Takeaway

Confirms mathematically that the visual shifts presented in previous slides are not random noise, but genuine, statistically significant ecological changes occurring in Similipal.

 Statistical rigour ensures that all observed trends are robust and reproducible findings.

Extracting and Visualizing Satellite Data via QGIS

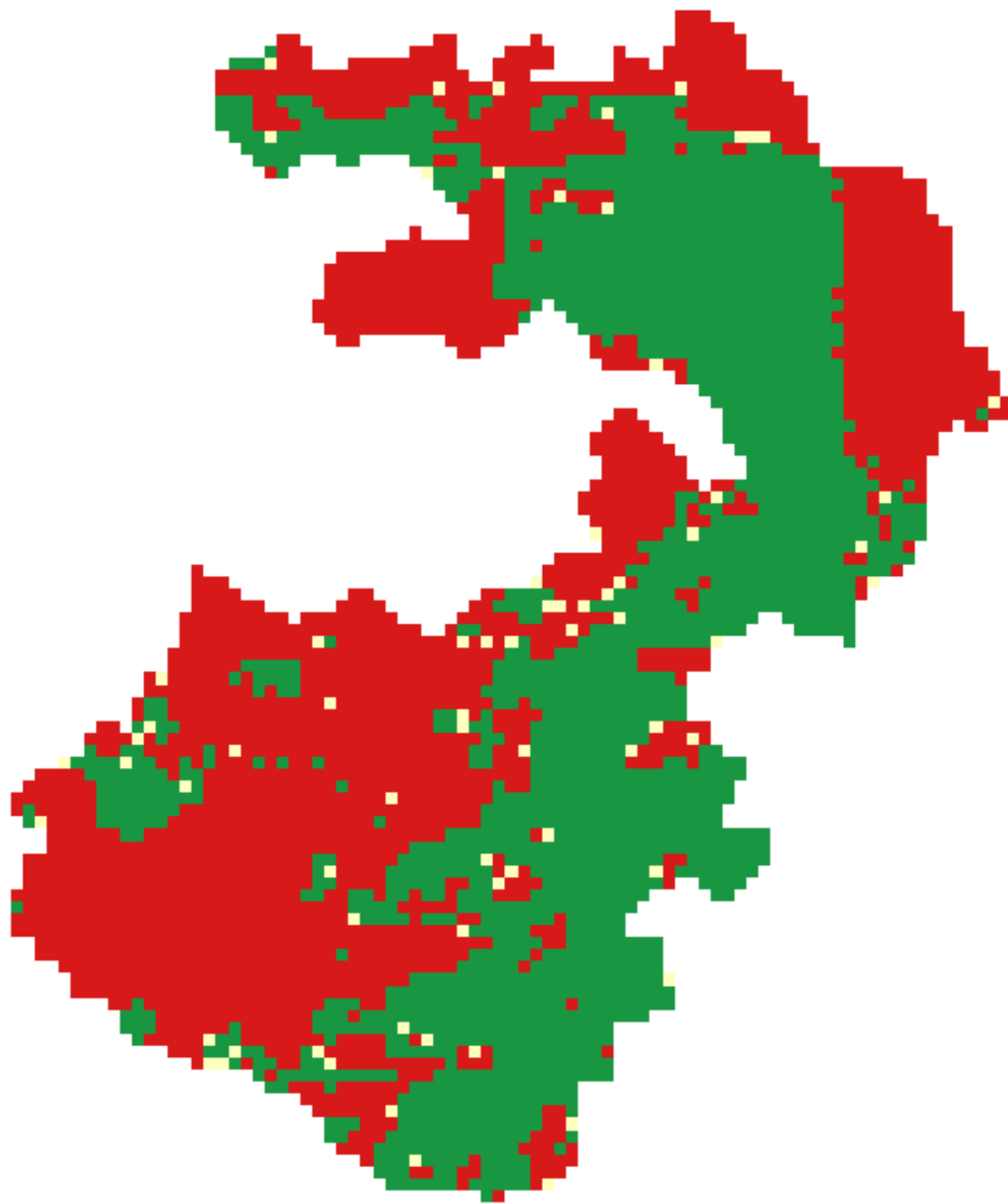
Raw remote sensing data (MODIS, CHIRPS) was extracted directly to capture essential climatic and phenological variables over Similipal. Temporal and spatial metrics (ET, LST, Rainfall, NDVI, Phenometrics) were converted into georeferenced raster image formats and imported into QGIS for multi-layered spatial visualisation, mapping out physical zones and variable intensities. This provides direct visual proof of how factors like moisture stress, surface temperatures, and rainfall correlate with the physical landscape of our Wet and Dry zones.



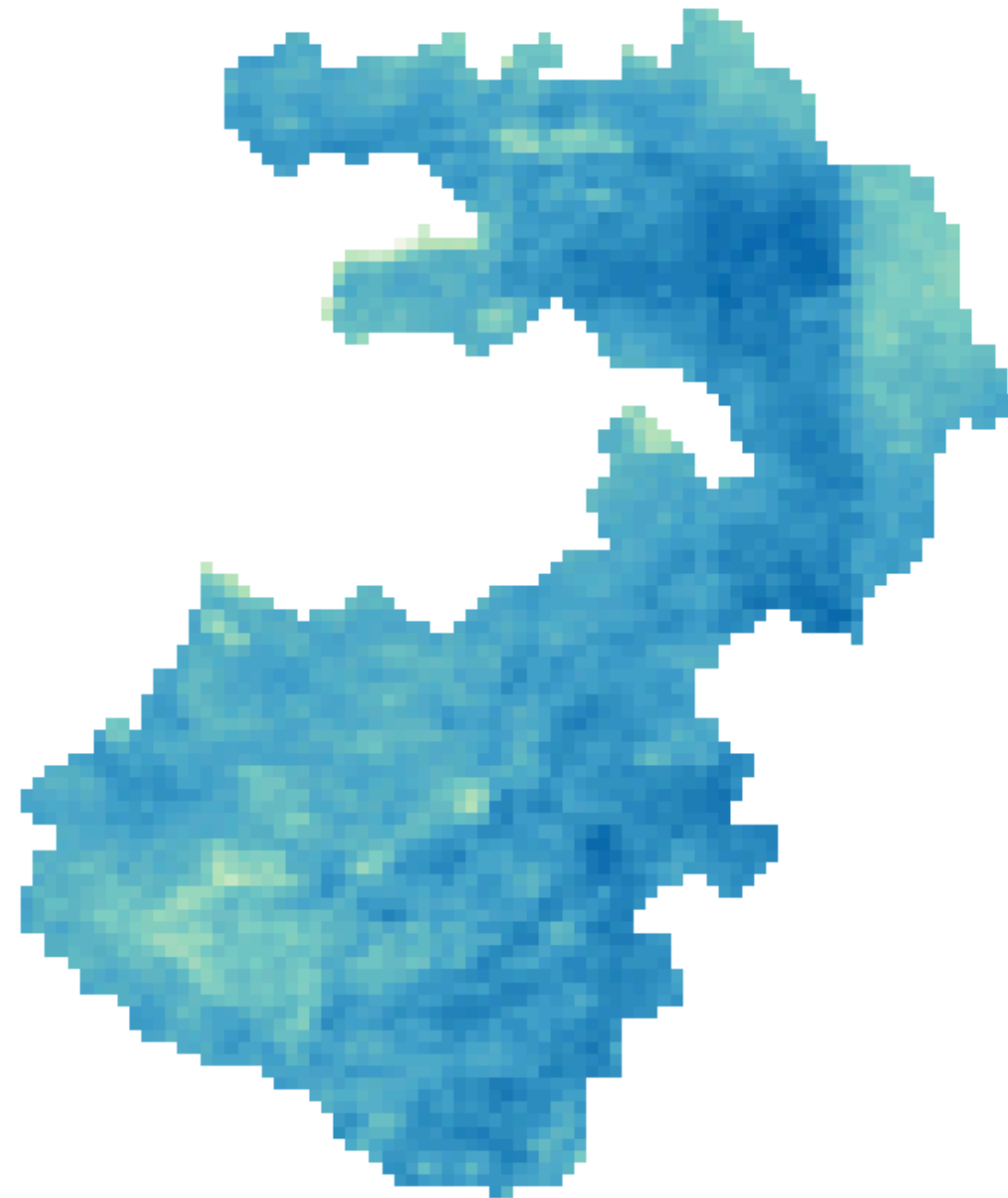
This workflow transforms raw satellite observations into spatially explicit evidence of climate–phenology relationships across Similipal's landscape.

Forest Zonation & Moisture Distribution

Spatial stratification of Similipal into Wet and Dry zones based on moisture gradients, providing the geographic framework for all subsequent analyses.



Zone Classification



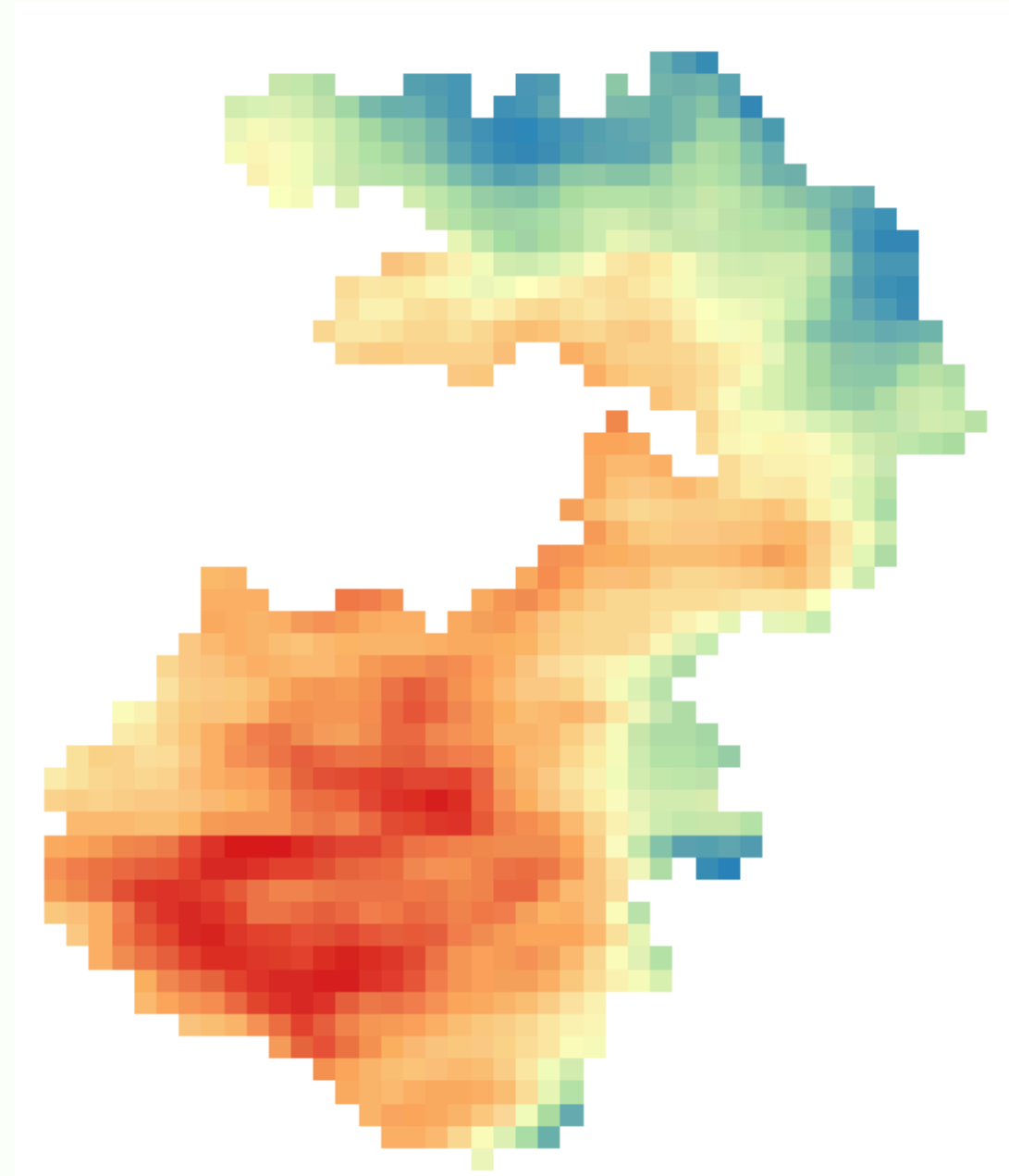
Moisture Map

Land Surface Temperature (LST) Patterns

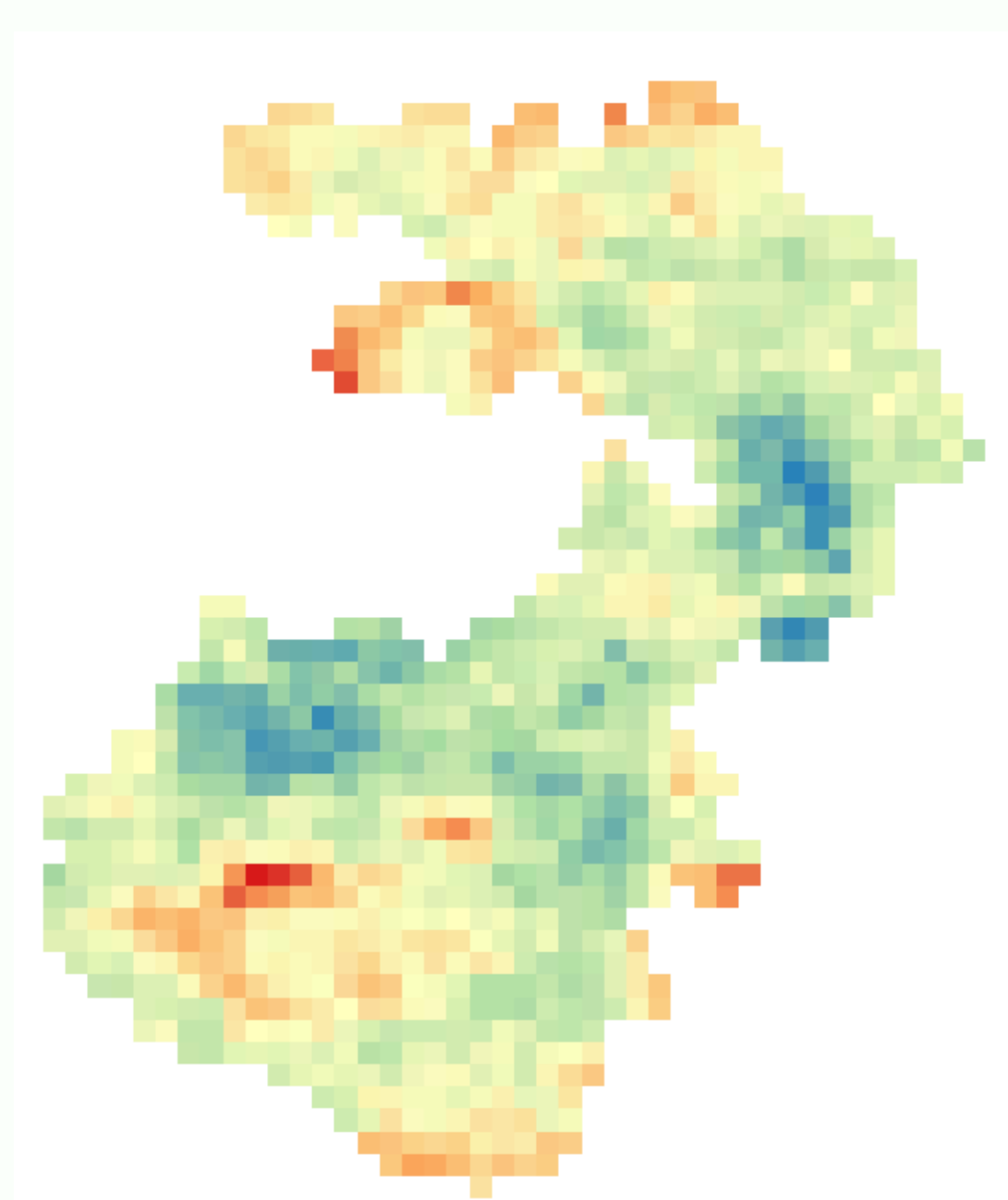
Thermal environment characterisation across Similipal, capturing daytime heating, nocturnal cooling, and long-term warming trends.



LST Day Mean



LST Night Mean



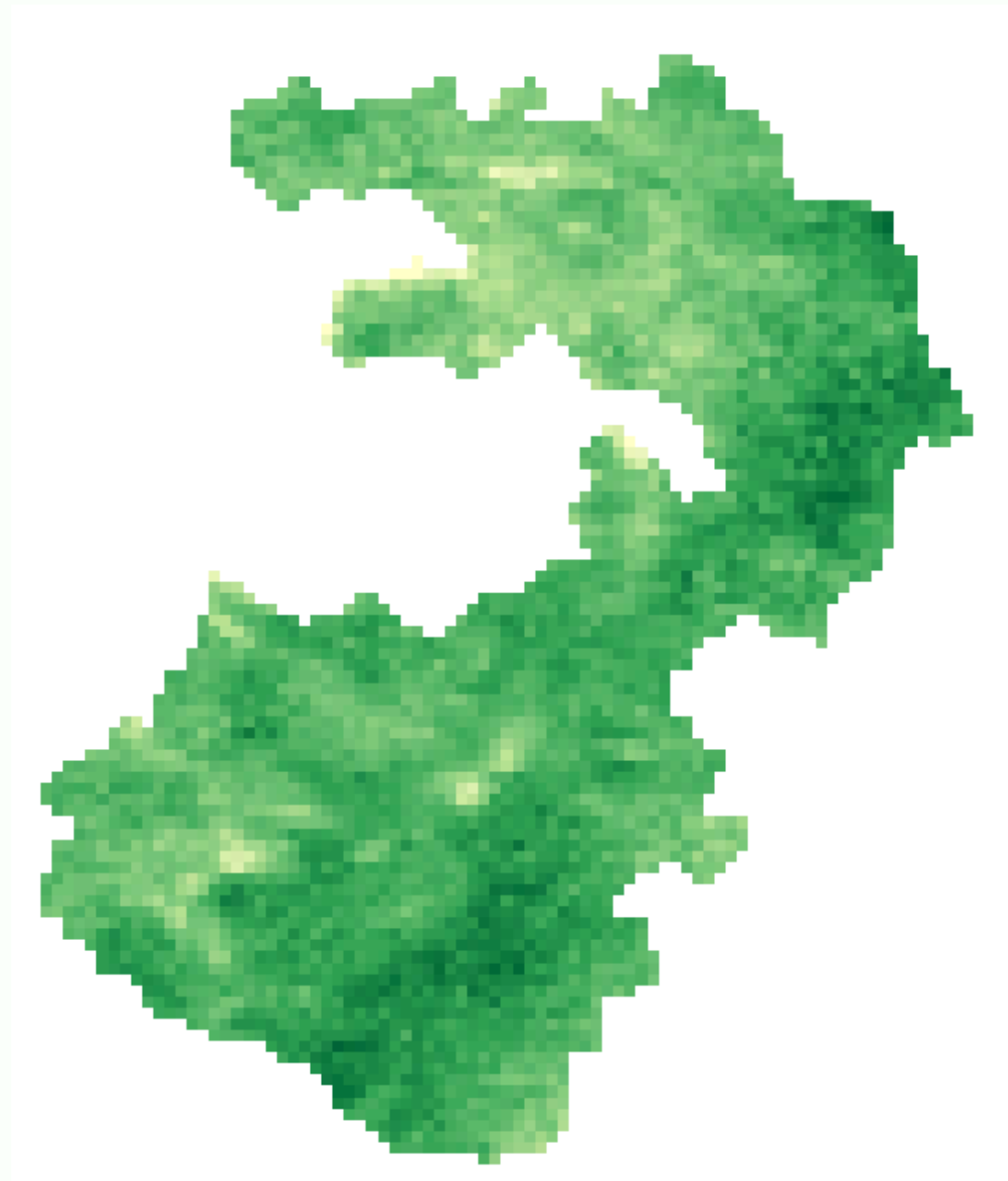
LST Trend

NDVI: 2001 vs. 2023 & Overall Trend

Vegetation greenness (NDVI) shifts over two decades, comparing baseline conditions in 2001 with recent state in 2023, alongside the overall trend surface.



NDVI 2001



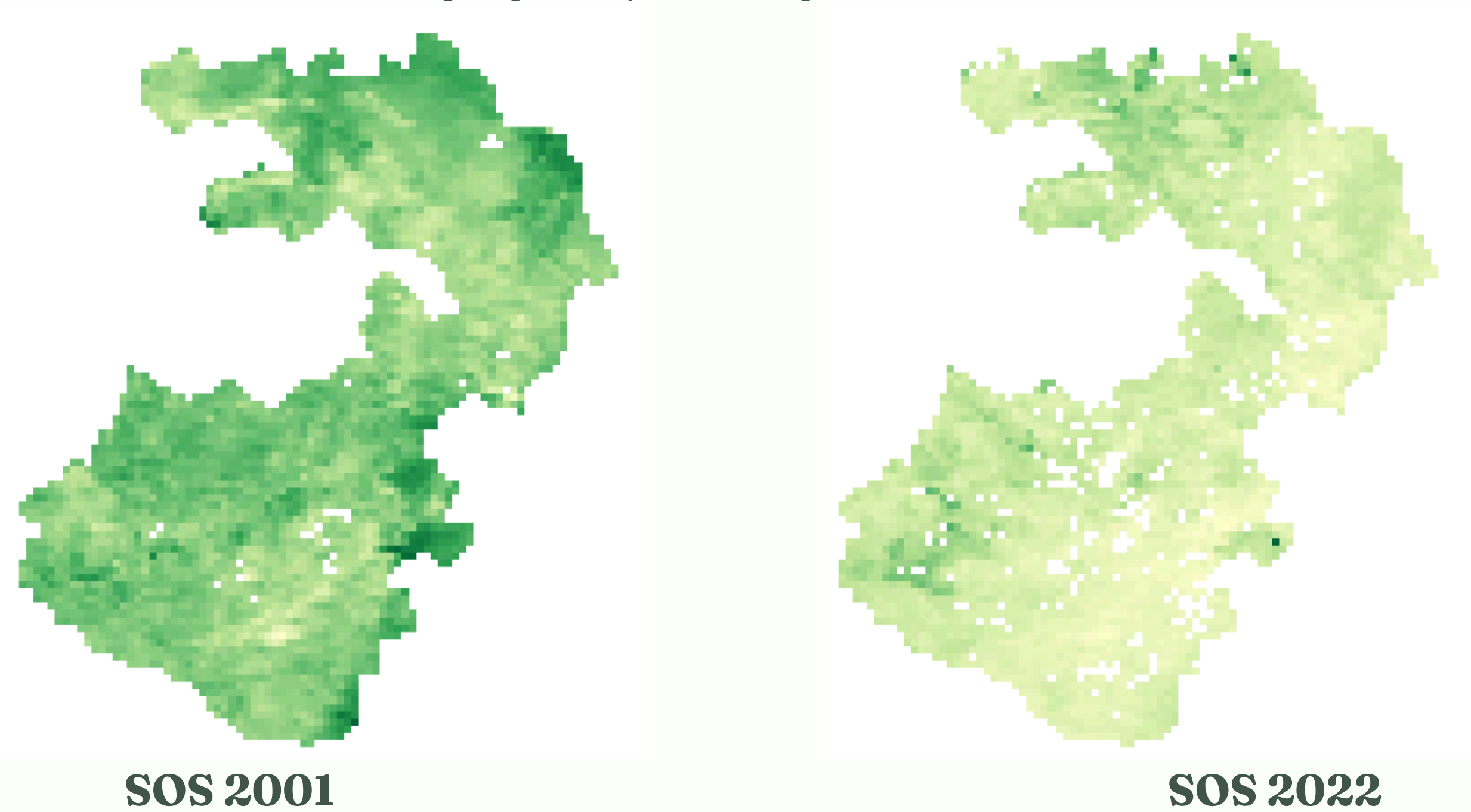
NDVI 2023



NDVI Trend

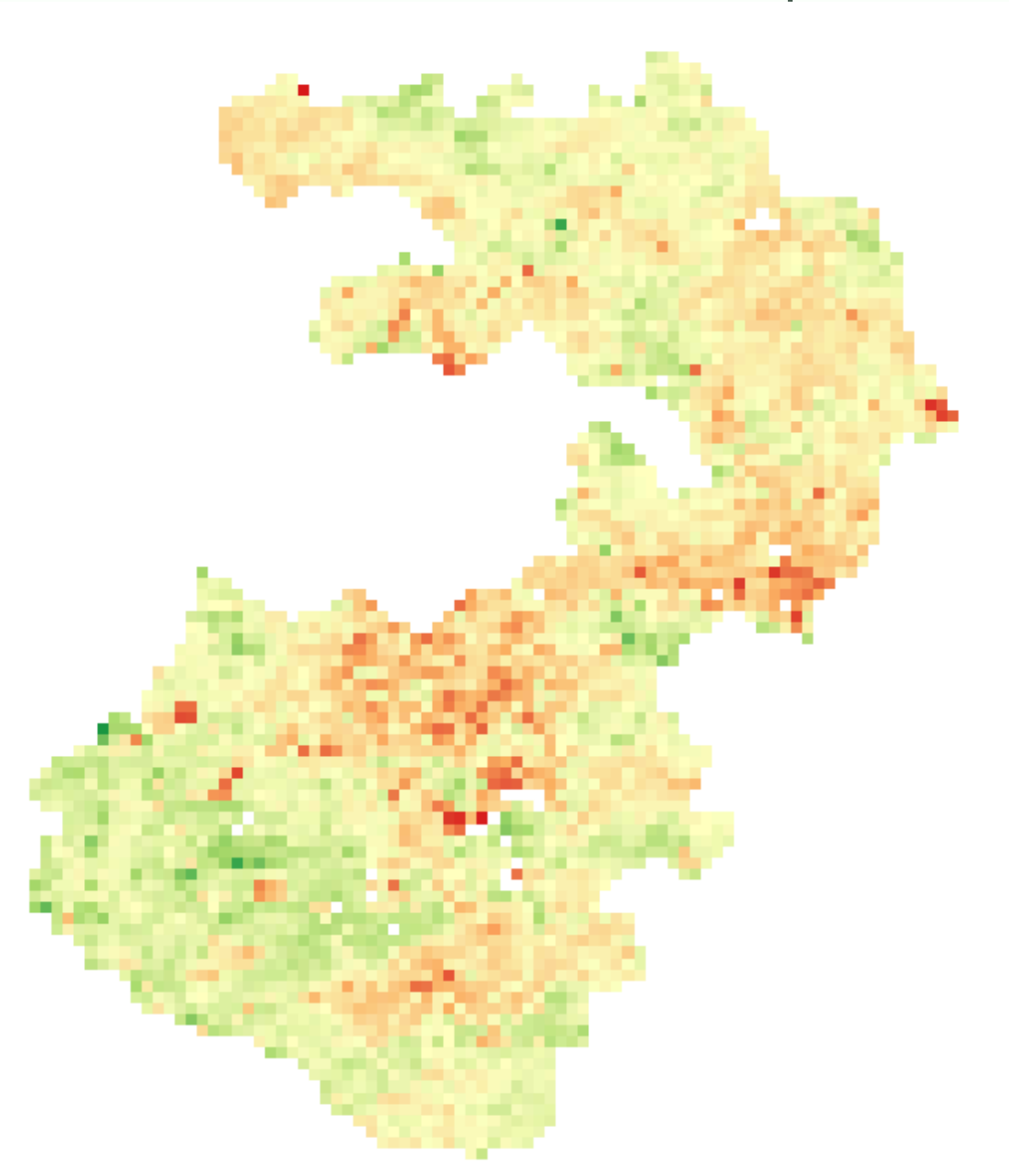
Comparing SOS Timing (2001 vs. 2022)

Spatial maps of Start of Season (SOS) reveal phenological shifts across Similipal, comparing the baseline year 2001 with 2022 to illustrate how the timing of green-up has changed over two decades.

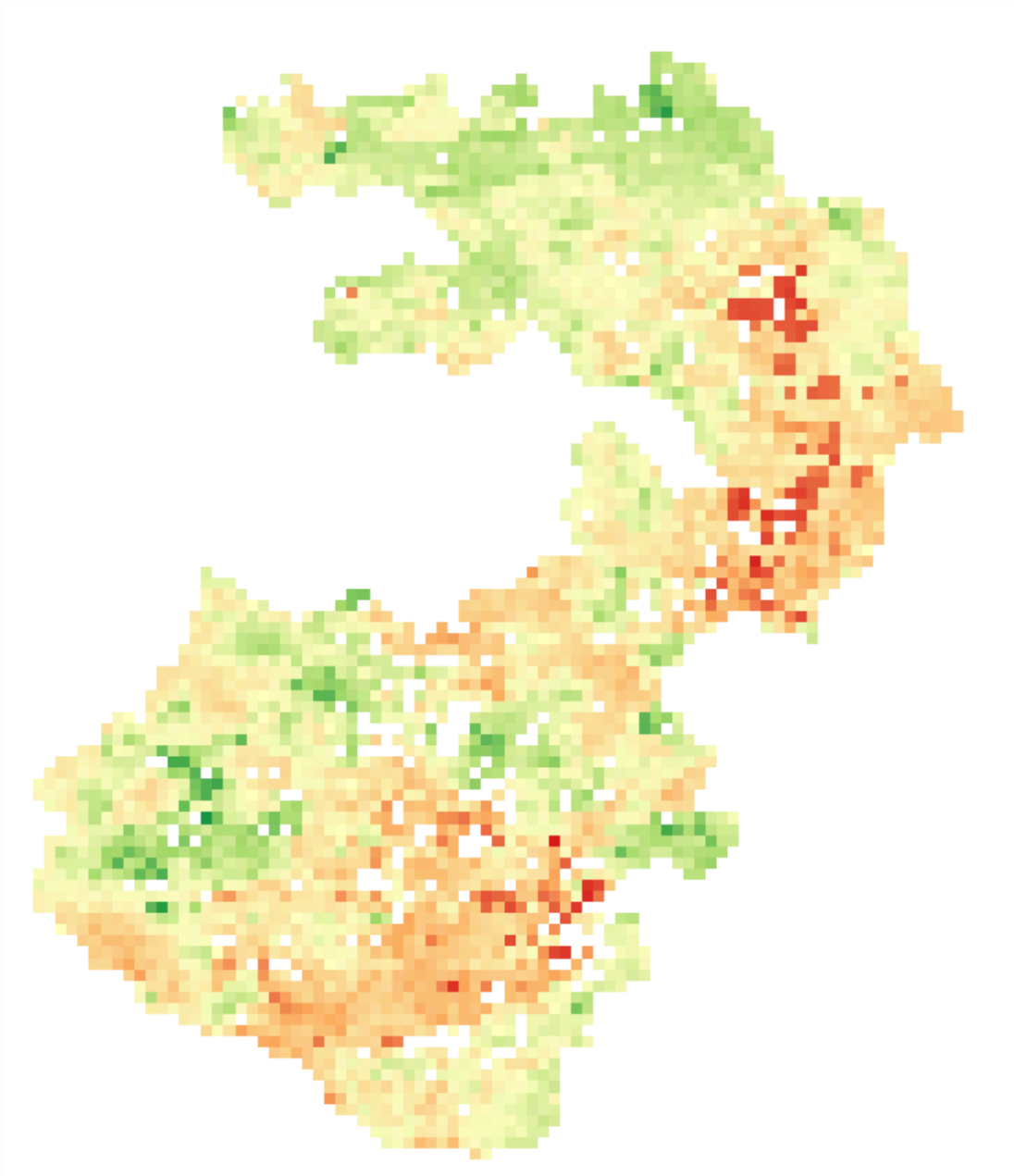


End of Season (EOS) & Season Length (LOS)

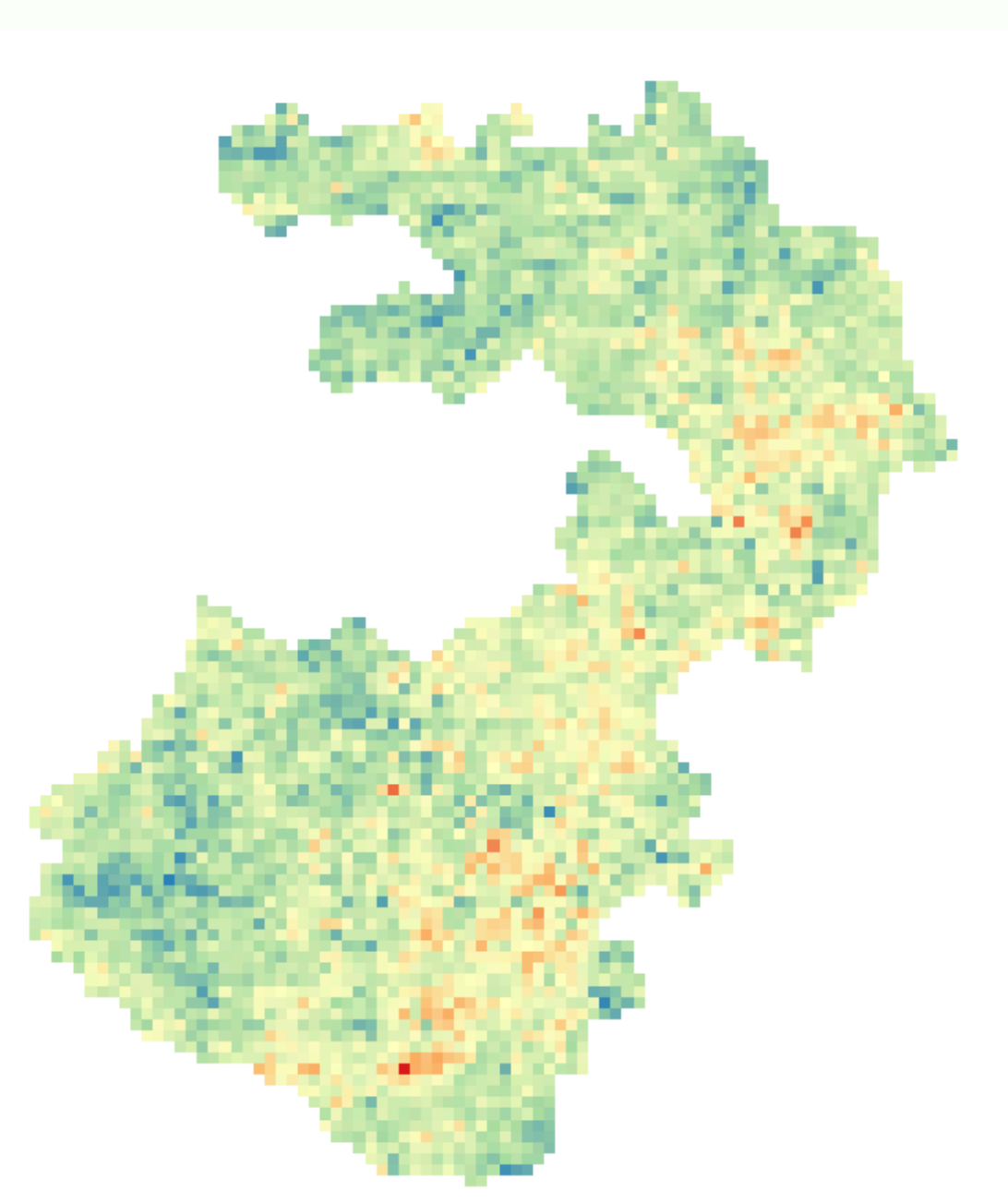
Maps of End of Season (EOS) for 2001 and 2022, alongside the Length of Season (LOS), complete the phenological characterisation of Similipal's forest dynamics.



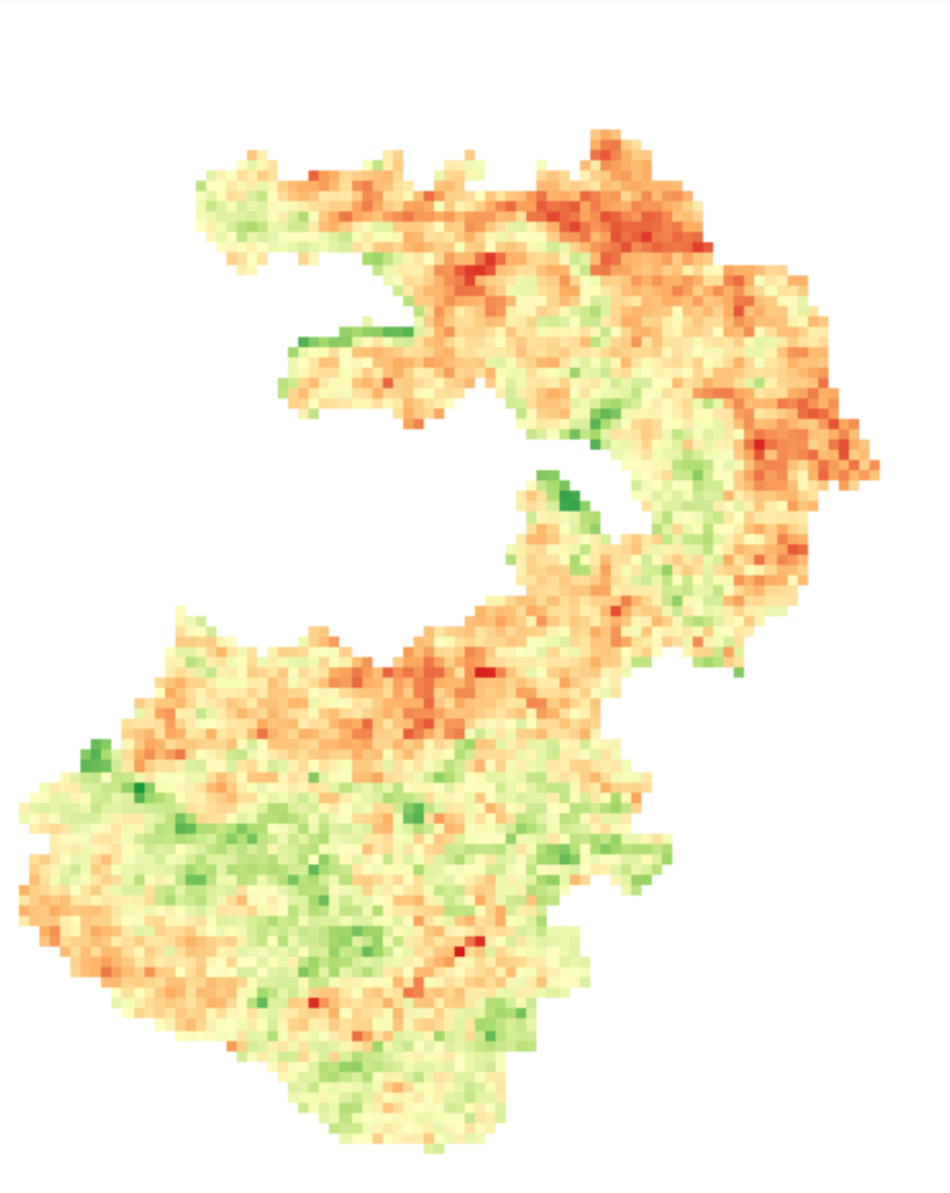
EOS 2001



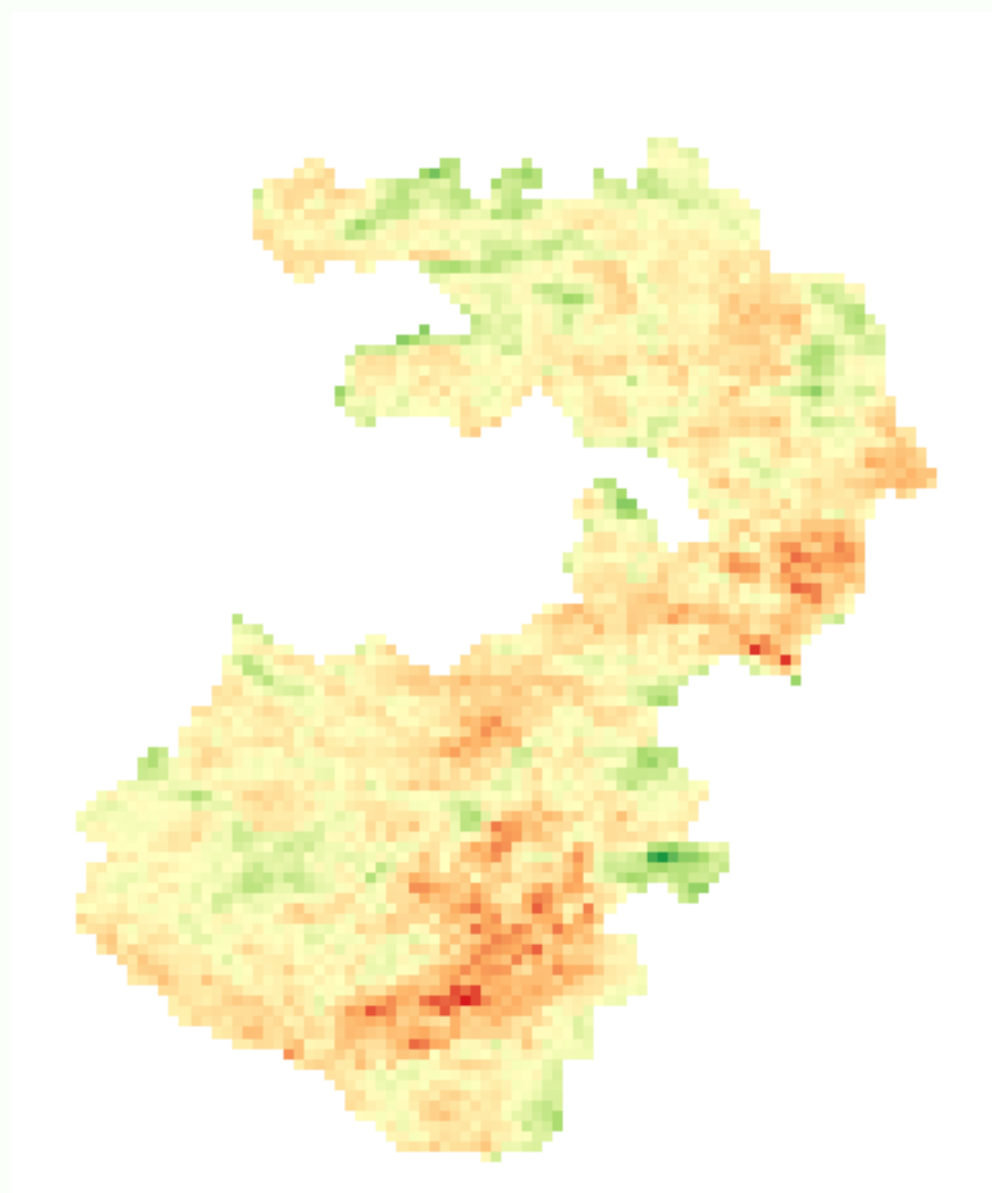
EOS 2022



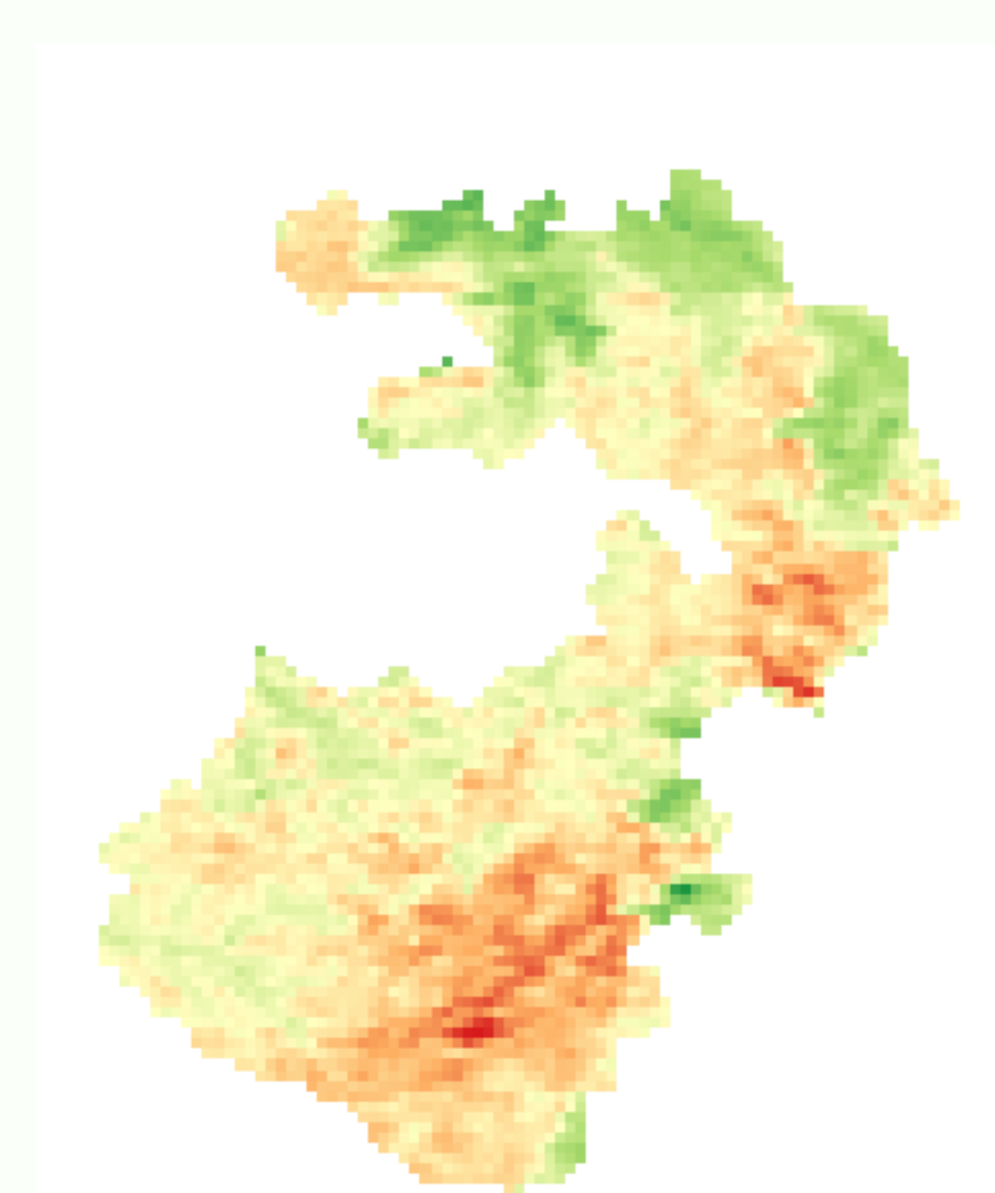
LOS (Length of Season)



Mean LOS



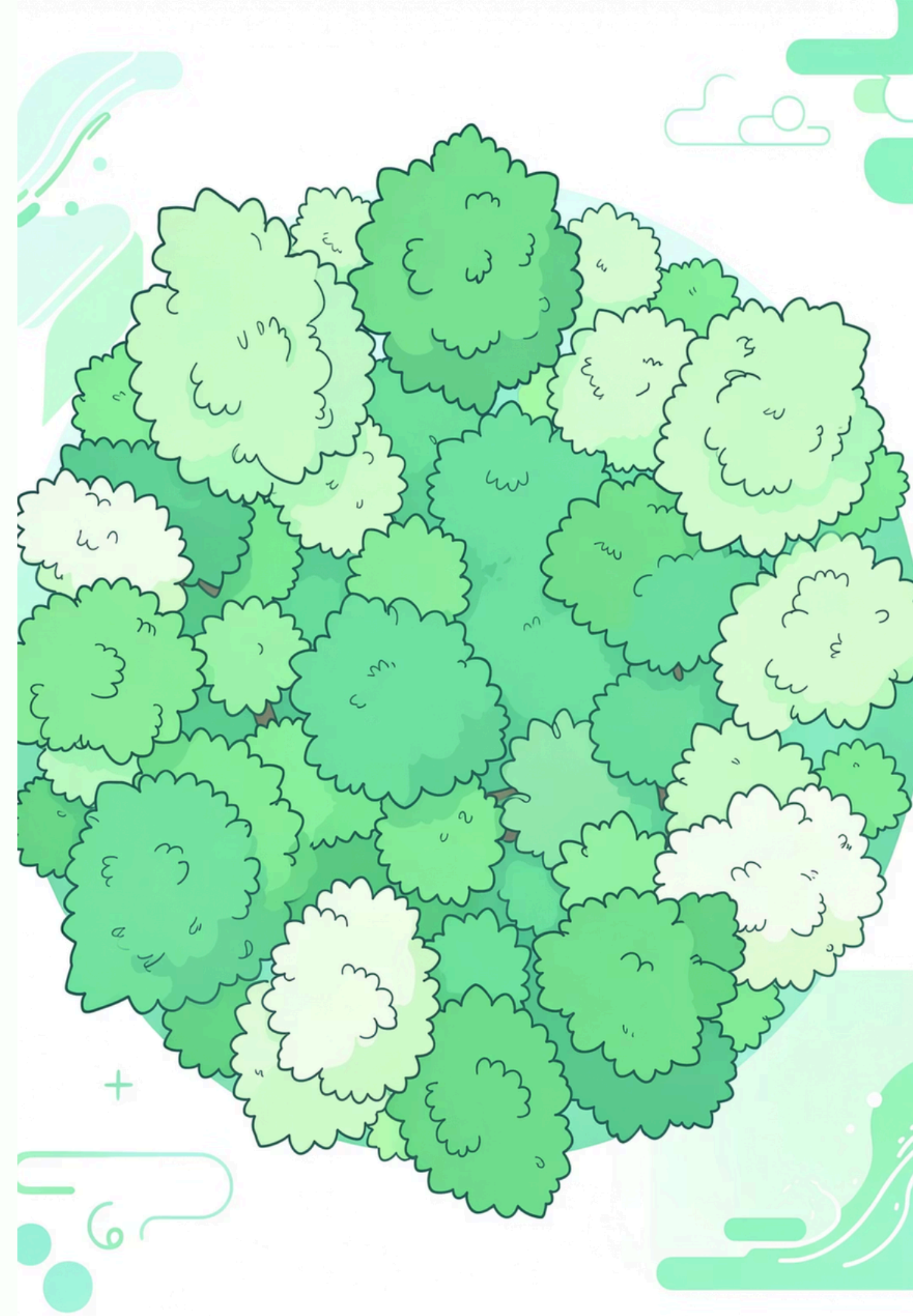
Mean EOS



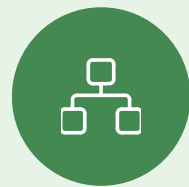
mean SOS

Machine Learning for NDVI Prediction

Literature Review: XGBoost & GWR Approaches



Research Paper 1: ML-Based NDVI Prediction



Model Architecture

XGBoost (Gradient Boosted Trees)

- Sequential learning
- Captures non-linear relationships



Climate Inputs

Temperature, rainfall, lag features



Output

Future NDVI prediction

- Vegetation dynamics
- Temporal patterns

Why this paper: ML for NDVI prediction using temporal + climate features

Research Paper 2: Climate–Vegetation Relationship

Geographically Weighted Regression

Captures spatial variation in NDVI–
climate relationships

- Regional differences
- Non-uniform responses

Key Drivers

Rainfall & temperature affect
vegetation differently across
regions



XGBoost Model Results

0.78

R² Score

Good prediction capability

90%

NDVI Lag

Dominant feature contribution

Model Performance

Captures NDVI trends through temporal patterns and climate influence

Random Forest Model Results



R² Score

Strong performance



RMSE

Low error

Slight improvement over XGBoost

Random Forest Model Results

0.4

R² Score

Medium prediction capability

50%

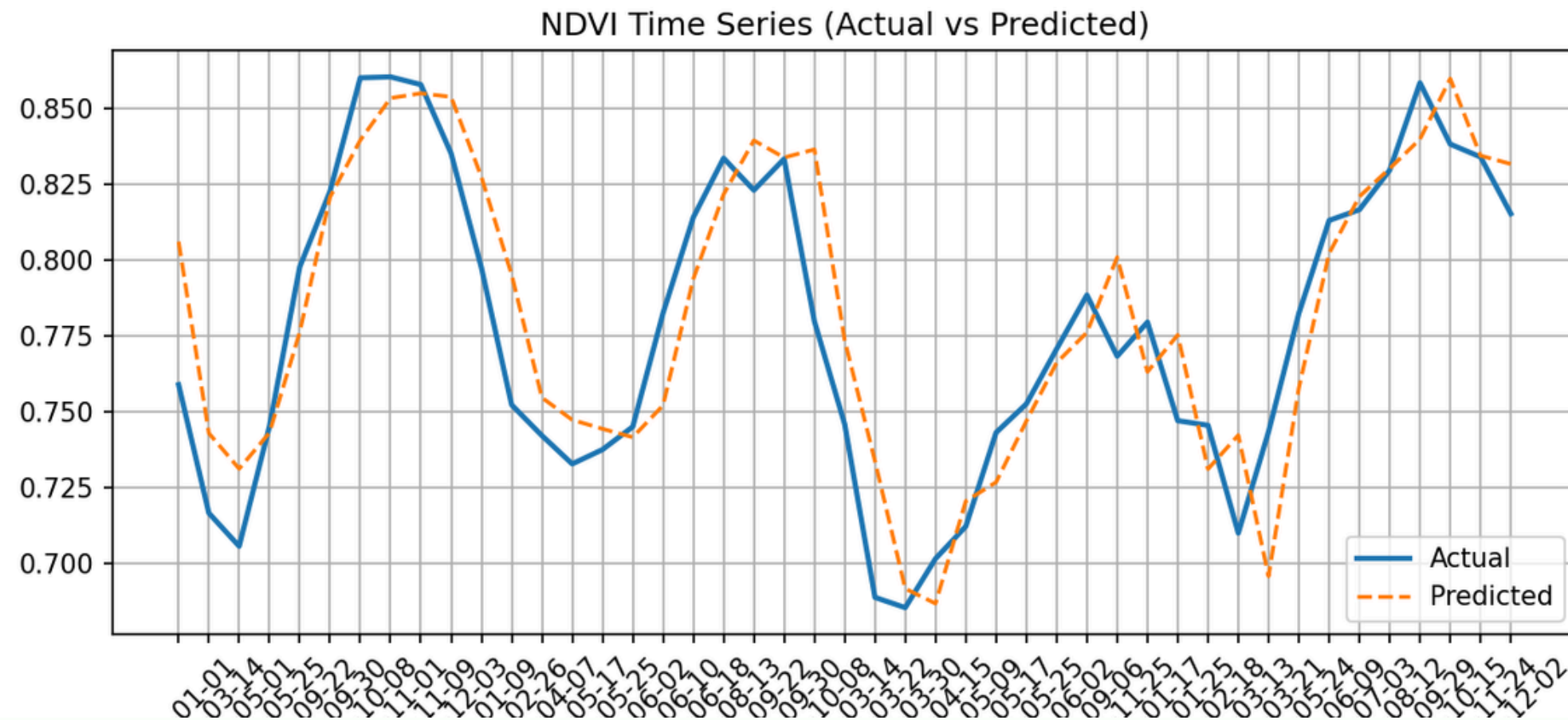
ET Stress

Dominant feature contribution

Model Performance

Captures NDVI trends through temporal patterns and climate influence

NDVI Prediction vs Actual (Time Series)



Key Observations

Predicted NDVI follows seasonal pattern

Peaks and drops correctly captured

- Strong alignment with actual NDVI

Enhancing Conservation in Similipal and Beyond

Our project provides critical insights and a robust framework for improving forest conservation efforts.

Key Takeaways for our AOI (Similipal):

- **Continuous Monitoring:** Acts as a continuous, non-invasive health monitor for Similipal's protected biosphere.
- **Targeted Interventions:** Identifies vulnerable Dry zones, helping management allocate resources earlier for wildfire prevention.
- **Early Warnings:** Uses LST and rainfall impacts as an early-warning system for climate-induced forest stress.

Broader Implications:

- **Scalable Framework:** Our GEE + Python + QGIS pipeline can be easily adapted to assess other protected forests in India.
- **Data-Driven Policy:** Shifts forest management from periodic visual surveys to continuous, empirical tracking for better climate resilience.



THANK YOU